



PANDASIA
PREVENTING FUTURE PANDEMICS



Relationship diagram for potential zoonotic spillover risks



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Pandemic literacy and viral zoonotic spillover risk at the frontline of disease emergence in Southeast Asia to improve pandemic preparedness

PANDASIA

Relationship diagram analyzing the complexity of spillover system and potential interventions

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2	NVI	Norwegian Veterinary Institute	Norway	WP4-Partner
3	UKHD	Universitätsklinikum Heidelberg University	Germany	WP5 lead
4	IZW	Leibniz Institute for Zoo and Wildlife Research	Germany	WP3 & WP6 partner
5	Chula	Chula University	Thailand	WP1 Lead
6	UMU	Umeå University	Sweden	WP4 lead
7	KKU	Khon Kaen University	Thailand	WP3 lead
8	MU	Mahidol University	Thailand	WP2 lead
9	SUPA71	SUPA71 Co., Ltd.	Thailand	WP6 lead
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List of Acronyms

CRD	Causal Relationship Diagram
3PL	Pandemic Prevention and Preparedness Literacy
QSDM	Qualitative System Dynamic Model
VHV	Village Health Volunteer
WP	Work Package
SAOs	Subdistrict Administrative Organizations
LGOs	Local Government Organizations
MOPH	Ministry of Public Health
NAPHS	National Action Plan for Health Security
IHR	International Health Regulations
SPS	Sanitary and Phytosanitary (measures)
FAO	Food and Agriculture Organization
FETN	Field Epidemiology Training Network
AI	Avian Influenza
MMR	Measles-Mumps-Rubella (vaccine)
IPC	Infection Prevention and Control
XB	Cross-Border
DLD	Department of Livestock Development
DNP	Department of National Parks
FGDs	Focus Group Discussions
CSO	Civil Society Organization
IHRI	Institute of HIV Research and Innovation
SHPHs	Subdistrict Health Promoting Hospitals
KAP	Knowledge, Attitudes, and Practices
PEP	Post-Exposure Prophylaxis
PODD	Participatory One Health Disease Detection

Executive Summary

Zoonotic spillover, the transmission of pathogens from animals to humans, is driven by interconnected factors such as deforestation, biodiversity loss, agricultural intensification, and climate change. These environmental disruptions weaken natural ecological barriers and heighten the likelihood of human-animal contact. In Southeast Asia, and particularly in Thailand's border regions, these risks are compounded by high-density interactions between people, livestock, and wildlife, unregulated cross-border trade, and limited veterinary infrastructure. Despite the cost-effectiveness of prevention, current policies tend to focus on reactive outbreak responses rather than addressing upstream drivers.

Community-level challenges such as low zoonotic spillover awareness, language diversity, and cultural practices also undermine prevention efforts. The One Health approach emphasizes coordinated action across human, animal, and environmental sectors. Thailand has embedded One Health into its national strategies and plays a leading role in regional networks and field epidemiology training. However, operational challenges remain, including siloed systems, limited local capacity, and fragmented policies. Strengthening community engagement, local governance, surveillance capacity, and intersectoral coordination is essential to mitigate zoonotic spillover risks.

Deliverable 9.1 aims to (1) map the zoonotic spillover system using causal relationship diagrams that reflect diverse stakeholder perspectives, (2) identify alternative One Health interventions beyond the 3PL intervention, highlighting their potential impact and the roles of community actors, and (3) support upcoming data collection by developing a ranked intervention matrix and a system visualization for stakeholder validation. This deliverable employed a qualitative, data-driven approach grounded in two focus group discussions (FGDs) with stakeholders from civil society and government agencies. These FGDs, conducted in October 2023, provided insights into local experiences, policy gaps, and zoonotic spillover drivers. Findings informed the development of a Qualitative System Dynamic Model (QSDM), visual tools, and an intervention ranking matrix for future stakeholder validation. Data from the FGDs were analyzed using causal relationship diagrams (CRDs) across five thematic areas and synthesized into thematic tables detailing proposed interventions. Ethical approval was granted, and participant confidentiality and anonymity were maintained throughout the study.

The deliverable identified five key themes contributing to zoonotic spillover risks through QSDM. First, proximity to land borders enables cross-border movement of animals and people, exacerbated by practices such as wildlife trade and unregulated livestock release. Second, environmental changes, particularly deforestation, forest encroachment, and climate-related flooding or drought, disturb ecosystems and increase exposure to waterborne and vector-borne diseases. Third, human mobility, including labor migration, tourism, and illegal crossings, complicates disease surveillance and access to healthcare, while abandoned animals and fear of legal consequences elevate transmission risks. Fourth, gaps in public health education and strong adherence to traditional beliefs hinder safe practices, with high-risk behaviors such as raw food consumption, poor hygiene, and inadequate pet vaccination contributing to exposure. Lastly, information technology plays a dual role: it supports remote access, data sharing, and

real-time communication, yet also faces limitations in infrastructure and the spread of misinformation. These findings emphasize the complex interplay of ecological, social, and behavioral factors influencing zoonotic disease emergence in Thailand's border regions.

The intervention matrix presents stakeholder-informed solutions across five themes. At the community level, measures focus on strengthening local resilience through financial support, public health education, and partnerships with trusted institutions like temples. For animal-related risks, stakeholders propose stricter regulations on pet ownership and wildlife interaction, improved inter-agency coordination, and national vaccination standards. Health system interventions emphasize enhancing the training and integration of Village Health Volunteers (VHVs), allocating resources for proactive surveillance, and reinforcing the legal frameworks guiding disease control. On policy and governance, the need for participatory lawmaking, clearer operational protocols, and decentralized, context-sensitive policy responses was underlined. Finally, leveraging tools, such as improved data sharing between research and health sectors, and expanded use of digital communication systems, was seen as essential for faster decision-making and better outbreak response.

This work expands WP1's initial framework into a mixed-methods approach, linking science and policy through context-sensitive, community-informed strategies. The QSDM and intervention framework provide a basis for the next stages of the project. These workshops will validate the system model, guide scenario-based modeling (WP4), and contribute to intervention co-design (WP5) and communication efforts (WP6). Future research could adapt and validate the model in other regions, such as Chanthaburi and Chiang Rai, using participatory methods to explore local dynamics and strengthen the model's relevance across diverse rural contexts.

Introduction

As part of Work Package 9 (WP9) from the PANDASIA project, NOVA is leading Task 9.1 with the goal of building a visual representation of the spillover system and identifying the most relevant and feasible One Health preventive interventions for zoonotic spillover, beyond the Pandemic Prevention and Preparedness Literacy (3PL) intervention. Our approach builds on insights generated by WP1 through stakeholder focus group discussions and contributes directly to PANDASIA's broader goals, specifically by identifying drivers of spillover risk (specific objective 1) and supporting the development of a public health measure to reduce transmission and pandemic risk (specific objective 7). Using stakeholders' insights, we synthesized a QSDM that visually represents links, actors, and processes contributing to zoonotic spillover in Thailand border regions. The QSDM provides a baseline visualization of key drivers and causal pathways of spillover, incorporating environmental, social, and behavioral dimensions. We also look at the existing policy structure and map how various actors, such as community members, local authorities, and policymakers, interact within the system and identify potential interventions across sectors.

The QSDM and interventions framework sets the foundation for upcoming activities. Future group workshops may use a Delphi method to collectively assess and generate a ranked matrix of One Health interventions based on feasibility, sustainability, and impact. In upcoming workshops, stakeholders will collectively evaluate and validate the spillover system by assessing the causal links within the QSDM, providing a shared understanding of the spillover system. This work will inform scenario-based modelling in WP4 by providing a visualization of the causal relationships within the spillover system. The One Health interventions identified by stakeholders will contribute to WP5 by giving insights into alternative or complementary interventions, and the spillover system will contribute to the co-creation of the 3PL intervention. We will support WP6 by sharing synthesized results to support communication and dissemination efforts. Through this work, NOVA expands the existing qualitative frameworks done by WP1 into a mixed-methods approach and supports PANDASIA's effort to connect science with policy through context-sensitive, community-informed intervention design.

Objectives

Deliverable 9.1 presents several relationship diagrams, identifies potential measures and interventions based on focus group discussions conducted by WP1, and informs the next data collections. These diagrams aim to represent stakeholders' perspectives from Thailand's border regions, regarding zoonotic spillover risks and the One Health approach. The deliverable has the following objectives:

1. Visualization of the spillover system through a QSDM, using a causal relationship diagram to synthesize multiple stakeholder perspectives on the links and processes of zoonotic spillover risks in specific scenarios, integrating environmental and social factors, community actors, habits, and behaviors.

2. Identify through QSDMs new potential interventions to the 3PL intervention based on the One Health approach and their potential impact to decrease the zoonotic spillover risk.
3. Provide a ranking matrix of the potential impact of identified interventions in terms of feasibility, sustainability, and potential impact, and by providing a visualization of the spillover system to be evaluated and validated by stakeholders in future data collection phases of the PANDASIA project.

Background

Zoonotic Spillover Risks and Prevention in Thai Border Regions: A One Health Perspective

Zoonotic spillover refers to the process by which pathogens are transmitted from animals to humans, often through a series of ecological and biological events, namely infection in wildlife, pathogen shedding, spillover to humans, and subsequent spread, often described as the "infect–shed–spill–spread" cascade [1], [2]. Literature from Southeast Asia suggests that zoonotic disease emergence and re-emergence, including rabies, highly pathogenic avian influenza (H5N1), and Nipah virus, can be influenced by multiple, interrelated factors, arising from interacting human–animal–environment drivers [3], [4], [5].

Research has linked spillover in Southeast Asia to a set of interacting human–animal–environment systems rather than isolated factors. For instance, environmental changes, such as deforestation, habitat fragmentation, and peri-urban encroachment, create more “edge” habitat, alter host-vector behavior, and erode landscape immunity, thereby increasing human–wildlife–vector contact, which can be linked to concrete events and trends such as the rise of zoonotic *Plasmodium knowlesi* malaria [6], [7], [8]. These phenomena have also been repeatedly linked to bat-borne zoonoses, including the shedding and spilling of pathogens such as the coronaviruses [1], [9]. Furthermore, droughts and landscape fires can alter wildlife movement and human exposure, as during the 1997–98 haze preceding the Nipah outbreak, and warming trends expanding climatic suitability for dengue and chikungunya transmission [10], [11].

Another factor of spillover risk is the food-system structure and related practices, such as backyard farming with poor biosecurity, high livestock density, land use patterns, and biodiversity loss [12]. For instance, studies have found that poultry farming consistently correlates with HPAI risk in Thailand and Vietnam [13], [14]. Live-bird markets in Vietnam are hosts for avian influenza viruses [15], [16]. Wildlife hunting, butchering, and trading increase shedding and cross-species transmission [17]. For instance, in the Laos market, 20% of traded wildlife tested positive for *Leptospira*, and poor hygiene magnifies exposure for traders and consumers [18], [19].

Despite the proven cost-effectiveness of preventive measures, global strategies largely remain focused on reactive preparedness and response rather than proactive prevention, a gap that can be addressed by following a One Health approach [20], [21]. Primary prevention at the source, such as deforestation reduction, smarter wildlife-trade governance, and targeted surveillance, has repeatedly shown to be far more cost-effective than reactive response post-spillover and carries substantial co-benefits for climate, biodiversity, and rural livelihoods [22], [23]. In parallel, targeted One Health surveillance that utilizes technology and risk-based sampling concentrated at wildlife–livestock–human interfaces and geographic hotspots identified through risk mapping, offers a pragmatic, cost-efficient way to intercept spillover early and guide rapid control, as recommended by recent reviews and empirical mapping studies [24], [25], [26].

Zoonotic spillover risks are suggested to be particularly elevated in Thailand’s border regions, where dense human-animal-environment interactions driven by agriculture, cross-

border trade, and migration may create favorable conditions for the emergence and transmission of infectious diseases [27]. Border regions like Chiang Rai, which is in northern Thailand and borders Myanmar and Laos and Chanthaburi, which is in eastern Thailand and borders Cambodia, face a diverse range of cross-border animal and human health risks [28], [29], introducing pathogens, exemplified by outbreaks such as the Nipah virus and avian influenza, as examples of potential consequences [30]. Despite these risks, these regions are described as facing major gaps in disease surveillance, harmonized reporting, veterinary infrastructure, and access to systematic vaccination and diagnostic programs, particularly in resource-limited rural areas [30], [31], [32].

The Role of Stakeholders and Community Actors

The One Health literature consistently identifies stakeholders from the human, animal, and environmental sectors who operate and interact across local, national, and international levels [5], [33], [34]. For instance, existing studies identify a wide range of stakeholders involved in the animal and animal product value chain, including community-level actors such as hunters, farmers, medical and veterinary practitioners, law enforcement and border patrol officers, and indigenous and local communities, including tribal and religious leaders and community representatives [35], [36], [37].

Beyond community and professional actors, non-governmental organizations (NGOs), government agencies and regulations, and international organizations are also reported to contribute to One Health efforts, which has led to calls in the literature for multisectoral coordination across local, national, and international levels in implementing One Health strategies [35], [38], [39], [40]. However, how these various actors coordinate in practice, and what roles they play under different governance structures or during specific public health events, remains unevenly documented, with studies noting fragmented networks, gaps at subnational levels, and persistent cross-border coordination challenges [41], [42].

Engaging stakeholders and community members can be an important component of zoonotic disease prevention, with roles in raising awareness, promoting behavior change, supporting early detection, and mobilizing local participation to enable culturally appropriate and sustainable One Health interventions and strategies [43], [44], [45]. In Thailand, the COVID-19 pandemic highlighted the importance of community actors in grassroots prevention and control efforts, as subdistrict health office personnel and over one million village health volunteers (VHVs) were reported to have supported 23 million households in more than 75 thousand villages, with home visits, risk screening, quarantine monitoring, hygiene education, and face mask production [46].

Thailand's risk communication strategy has relied on combining traditional media, such as TV, radio, and newspapers, and digital platforms, such as Line, Twitter, Facebook, and TikTok, to deliver targeted health messages and promote behavioral change [47]. Furthermore, a two-way communication with stakeholders and influencers is described as an approach that may help ensure that messages are accessible in multiple languages and formats, reaching

diverse groups including migrant workers, vulnerable populations, and individuals with visual impairments [48], [49]. For instance, national campaigns such as the CCSA and ThaiHealth “Thai Roo Soo COVID” effort used multi-channel outreach and encouraged more interactive, two-way communication with the public [50]. To enhance this approach, sources emphasize the need to recruit and train culturally and linguistically competent spokespersons and volunteers. In Thailand, this has included interpreter-mediated migrant health worker/volunteer programs and cultural mediation services for Burmese, Cambodian, and Lao speakers, alongside risk-communication initiatives tailored to migrant workers and accessible formats for people with visual impairments [51], [52], [53]. VHV, community leaders, and village heads have been reported to play a role in proactive disease surveillance and in community engagement, especially among ethnic minorities and remote border populations, where misinformation and cultural beliefs contribute to vaccine hesitancy and where access to health services is often limited [48], [54], [55].

Local Community Interventions

Preventing zoonotic spillover benefits from actively involving communities in identifying risky behaviors, such as unsafe animal handling or poor hygiene, and promoting safer practices through locally appropriate education and discussion [56]. Giving communities a voice in planning and decision-making may help build trust and encourage long-term behavior changes that reduce disease risks at the human–animal–environment interface [57]. Shared planning and decision-making with communities, through co-creation and co-design with local leaders, Indigenous groups, and practitioners, builds trust and yields culturally appropriate, context-specific interventions that are more likely to be adopted and sustained [58], [59]. These approaches are especially critical for groups at the front lines of the human–animal–environment interface in Southeast Asia, such as smallholder farmers, forest-going ethnic communities, and border populations, who raise livestock, forage or hunt wildlife, and engage in cross-border market chains that bring them into frequent contact with pathogen hosts and vectors [60], [61].

In the Thai context, several examples illustrate how culture, interface risk, and multisector coordination shape zoonotic-risk management, from ethnographic risk patterns to community education and integrated One Health control programs. For instance, ethnographic work in Mukdahan’s Phu Thai community underscores how a community that maintains ancestral practices such as the consumption and ceremonial use of raw meat from wild animals, believes in spiritual causes of illness, and relies on traditional healers, was able to adapt to the COVID-19 pandemic era and improve health awareness [62]. In Chonburi’s Wat Luang village, home to Thailand’s largest roost of *Pteropus lylei* bats and located near pig farms and human dwellings, community members were trained using culturally adapted materials like the ‘Living Safely with Bats’ booklet to raise awareness of Nipah virus risks, as part of a One Health intervention aimed at reducing spillover through improved understanding of human-bat-

pig interactions [63]. This initiative helped reduce risky behaviors, enhance vigilance, and build preparedness at the local level [63].

In Chanthaburi, a knowledge, attitudes, and practices (KAP) survey linked knowledge gaps about rabies and risky practices such as letting dogs roam unvaccinated and using traditional remedies, to targeted health education campaigns and community surveillance to improve awareness, promote dog vaccination, and ensure access to post-exposure prophylaxis (PEP) [64]. The Lawa Model implemented in Lawa Lake communities in Khon Kaen demonstrates how integrated community-based programs can sustain control of an endemic zoonosis such as the liver fluke infection (*Opisthorchis viverrini*), through health education for students and villagers, screening and treatment for infected individuals, surveillance and control of parasite hosts, and environmental interventions to reduce transmission [65], [66].

Together, these case studies suggest that local community interventions may contribute to zoonotic disease prevention by improving awareness, surveillance, and multisectoral coordination at human–animal–environment interfaces. [67], [68]. They also highlight key barriers to community engagement in zoonotic disease prevention, such as cultural practices that may conflict with health recommendations, lack of trust in authorities or health workers, communication difficulties due to language diversity, and low knowledge and literacy levels [65], [69]. Another major challenge is the lack of resources, including insufficient numbers of trained facilitators to work at the community level, limited access to educational materials tailored to local contexts, inadequate transportation and logistical support for outreach activities, and a lack of sustained funding to support long-term engagement and follow-up, and a weak coordination across human, animal, and environmental health sectors [65], [70].

Technology use for prevention

Recent initiatives in Thailand have been increasingly pairing digital tools with a participatory One Health approach to enable early detection and timely outbreak control. For example, a system in Chiang Mai province called Participatory One Health Disease Detection (PODD) mobilizes community volunteers to monitor and report animal, human, and environmental problems via a mobile app that automatically alerts authorities, which resulted in rapidly containing multiple livestock outbreaks before wider spread occurred [71]. Thailand's avian influenza (AI) surveillance system also utilizes local communities through mechanisms such as village health volunteers (VHVs) and farmer-reported mobile applications, particularly in border provinces with high poultry density and migratory bird activity, where human-animal-environment interactions are frequent [72].

VHVs have emerged as key users of these digital channels, integrating mobile applications such as the SMART VHV platform into their workflows, which was reported to improve the delivery of public health services, especially in underserved areas [73]. For instance, the OSOMO prompt mobile application was described as supporting VHVs in rural Thailand by streamlining elderly care through functions like health assessments, home visits,

and emergency reporting, which was reported to improve work efficiency, with users rating it highly for ease of use and feasibility in enhancing community health services [74]. Another example is the “Stroke Sign by Sticker” in LINE, which was presented as an effective and user-friendly tool for improving stroke knowledge among VHVs, which was reported to improve VHVs’ ability to recognize stroke symptoms, potentially contributing to better community preparedness for stroke care and education [75].

Methods

Study Design

The study design followed a data-driven approach, starting with qualitative data collection through FGDs with stakeholders from NGOs and government agencies. These discussions provided valuable insights into community-level experiences, policy gaps, and systemic drivers of zoonotic spillover. Drawing on this input, we developed the QSDM that identifies causal relationships focused on major topics of concern, as well as an initial mapping of potential One Health interventions that go beyond the 3PL intervention. The process culminates in the development of practical tools for the next stage of data collection, including an intervention ranking matrix and visual materials for stakeholder validation. This ensures that future workshops and scenario modeling are grounded in locally informed system insights and that proposed interventions reflect the realities and capacities of communities in Thailand's border regions. Figure 1 illustrates this design.

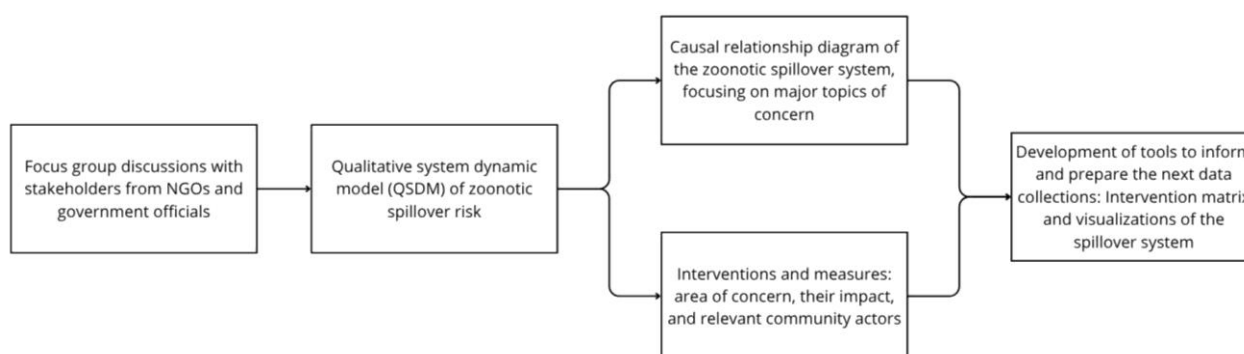


Figure 1. Study Design

Data Collection

The study drew on two FGDs with purposively selected participants representing key stakeholder groups involved in zoonotic disease prevention and response. The focus group discussions were conducted in October 2023, in collaboration with WP1. The first group integrated representatives from local civil society organizations (CSO group), who encompass the roles of working directly in civil society and with communities, informing communities about changes in the law, guiding communities in checking legal updates online, mediating when communities disagree with legal changes and building knowledge-based resilience in communities. The first group had representatives from the Zoological Society of London, Seub Nakhasathien Foundation, World Vision, and the Institute of HIV Research and Innovation (IHRI), as well as professionals involved in research communication, mass communication, and community engagement with local leaders.

The second group consisted of government stakeholders, including staff from the Provincial Public Health Office responsible for communicable disease oversight, strategic planning, surveillance, and policy implementation. Additional participants included

veterinarians from the Department of Disease Control and the Provincial Livestock Office, focusing on zoonotic surveillance and coordination for diseases such as rabies and avian influenza. Also present were frontline health workers from Subdistrict Health Promoting Hospitals (SHPHs), tasked with local disease prevention and education, as well as district-level public health officers overseeing outbreak investigations, resource management, and disease monitoring.

Ethical Considerations

This study adheres to KKU's ethical guidelines. Participants were provided with comprehensive written information about the study before giving informed consent in written form. Confidentiality and anonymity are strictly maintained throughout the research. All data collected is anonymous and will be analyzed and reported in an aggregated form to ensure that no individual participant can be identified. No personal or demographic details, such as names, age, and gender, are shared and have been retracted from the transcriptions and translations. Emphasizing anonymity is intended to encourage honest and candid responses [76]. Measures will be taken to uphold participant privacy and address any potential ethical concerns related to sampling and data handling.

Data analysis: Causal Relationship Diagrams (CRDs)

For the qualitative analysis in this study, data was extracted from two FGDs. Participant statements were first categorized by topic, after which we identified the key variables and the causal links described within each statement. These variables and relationships form the basis for constructing causal relationship diagrams (CRDs), which visually represent how different elements interact within the system [77], [78]. To organize the analysis, we grouped the data into five major thematic areas related to zoonotic spillovers: (1) animals and land border proximity, (2) environmental issues and changes, (3) movement of people, (4) knowledge and beliefs, and (5) use of information technology. A separate CRD was developed for each theme to capture the specific causal dynamics within each area.

Data analysis: Potential interventions

We analyzed stakeholder statements to identify potential, new, and alternative interventions across multiple sectors. We aimed to uncover context-specific and cross-sectoral solutions that address the interconnected challenges of human, animal, and environmental health. Drawing from direct quotes, we systematically answered four guiding questions: (1) What problem is being identified, (2) What measure or intervention is suggested, (3) How and by whom it can be implemented, (4) What is the expected impact, and on whom.

We organized these findings into thematic tables under five key topics: (1) potential interventions at the community level, (2) potential interventions covering animal interaction, (3) health-focused potential interventions, (4) potential interventions related to policy making and governance, and (5) potential interventions related to leveraging tools. Each table presents the problem, potential intervention, implementation strategy, and expected impact alongside a supporting quote, making the logic and framing behind each suggestion visible. In doing so, we aimed not only to document stakeholder-informed measures but also to uncover the relationships and points of collaboration among national and local actors.

Understanding causal relationship diagram building

CRDs are a foundational tool in system thinking and system dynamics for representing the feedback structure of complex systems. A CRD is a visual map of a system's core elements and behavior emerging from their causal interactions [79]. By depicting chains of cause-and-effect influence, CRDs help analysts capture the dynamic interactions that produce growth, decay, or equilibrium in systems. A CRD consists of variables connected by directed arrows that denote causal influence. Each causal link is assigned a polarity, indicating whether a change in the cause produces a change in the effect in the same or opposite direction [80]. A positive link (+) indicates that the change occurs in the same direction. Specifically, an increase in the cause leads to an increase in the effect, while a decrease in the cause results in a decrease in the effect (Figure 1).

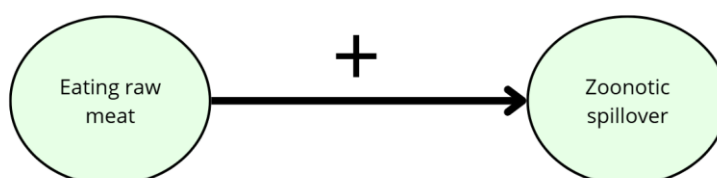


Figure 2. Positive link

Conversely, a negative link (-) represents the opposite reaction. This means that an increase in the cause results in a decrease in the effect, while a decrease in the cause results in an increase in the effect (Figure 2).

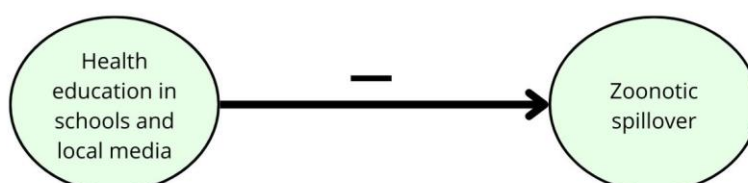


Figure 3. Negative link

Results: Qualitative system dynamic model (QSDM)

Relationship diagram building for major topics

The QSDM is divided into five CRDs, each representing a major topic. The figures below illustrate the key pathways identified through qualitative analysis that contribute to the risk of zoonotic spillover in the studied context. The relationships depicted are grounded in participant perspectives, and the supporting quotes said by stakeholders from NGOs (FGD 1) and government officials (FGD 2) are presented in the accompanying table.

Topic 1: Animals and land border proximity

A central concern voiced by participants was the risk posed by cross-border animal movement, especially in areas where the landscape allows for unrestricted travel. Proximity to international borders, in this case with porous natural boundaries, was perceived as a major driver of disease introduction. One respondent described the situation as follows: *"There is a risk from animals that may come from outside the country... Being near the border, animals or people can cross over easily. There is no real boundary, just a mountain that can be crossed on foot. (FGD 2)"* This context facilitates the movement of wild and domestic animals across borders, increasing the likelihood of contact with unfamiliar pathogens and accelerating potential zoonotic spillover.

The hunting and trade of wild animals, especially near border areas, was also identified as a risk-enhancing practice. In some cases, wild species such as bats are captured, consumed, or traded internationally, which intensifies human-wildlife contact. *"It's [bats] made into a dish called 'Tom Phet.' Mostly, it gets sent abroad to Vietnam. (FGD 1)"* Such practices, while culturally embedded or economically driven, elevate the probability of pathogen transmission through handling, slaughter, or consumption, and are directly linked to the emergence of zoonotic diseases. Compounding the issue, cattle smuggled across borders, such as from Myanmar, may bypass health checks and are deliberately released to avoid inspection. *"Sometimes they try to hide, like cattle from Myanmar, they evade officials and get released. (FGD 1)"* This practice creates significant gaps in animal health surveillance, contributing to uncontrolled disease transmission risks. Moreover, these forest areas may also be sites of armed poaching activity, as described by participants, further complicating efforts to monitor and manage animal movement.

Additionally, participants described specific practices related to cattle management along the border, which further illustrate how informal systems can pose health risks. In some areas, cattle are intentionally released into the forest, either for grazing or to evade detection when smuggled across borders. *"Like cattle being released into the forest... They use a method where they sprinkle salt every 7 days, and they set up a pen there to bring them back. (FGD 1)"* This salt-baiting technique allows herders to retrieve free-roaming cattle without fencing or close supervision. However, such livestock may remain in contact with wildlife for extended

periods, increasing exposure to pathogens. *"The issue is that pets live with people, and communities within the parks, such as wild buffalo and domesticated buffalo, may crossbreed. (FGD 1)"*

Another major driver of exposure arises from community proximity to forests and protected areas, which increases the likelihood of both direct and indirect contact with infected environments. For example, tourism or daily forest use can lead to unintended interactions with contaminated natural spaces: *"Tourists might enter forests and unknowingly come into contact with infected areas, such as trees with hollows. (FGD 2)"* In these cases, the forest itself becomes a reservoir for pathogens, and contact can occur without direct interaction with wildlife [81].

This proximity also encourages shared use of natural resources, especially water sources, between wild and domestic animals. Participants expressed concern about crossbreeding between wildlife and livestock, particularly red cattle and buffalo, which may facilitate the transmission of infectious diseases between species. *"There are over 20 of them (red cattle) now, and there is a possibility of crossbreeding with domestic cattle, which could introduce new diseases because they share the same water sources. (FGD 1)"* Such interspecies interactions, especially in unmanaged settings, present clear spillover risks as they break down the barriers between wildlife reservoirs and domestic hosts.

Additionally, food availability near forest edges was found to influence wildlife behavior, encouraging species to inhabit areas closer to villages and farmland. *"The species (of elephants) that reproduce will stay at the forest's edge because they have a full food supply. (FGD 1)"*

This ecological dynamic increases the presence of wild animals outside protected areas, which in turn presents management challenges for local and national authorities. To address this, some respondents mentioned the use of sterilization programs aimed at reducing wildlife populations outside conservation zones for elephants and monkeys: *"There are measures such as sterilization to reduce the population. (FGD 1)"* These efforts are represented in the diagram as negatively associated with risk, since reducing wildlife numbers and movement in human-occupied areas could lower the frequency of human-wildlife encounters and, consequently, reduce disease transmission opportunities.

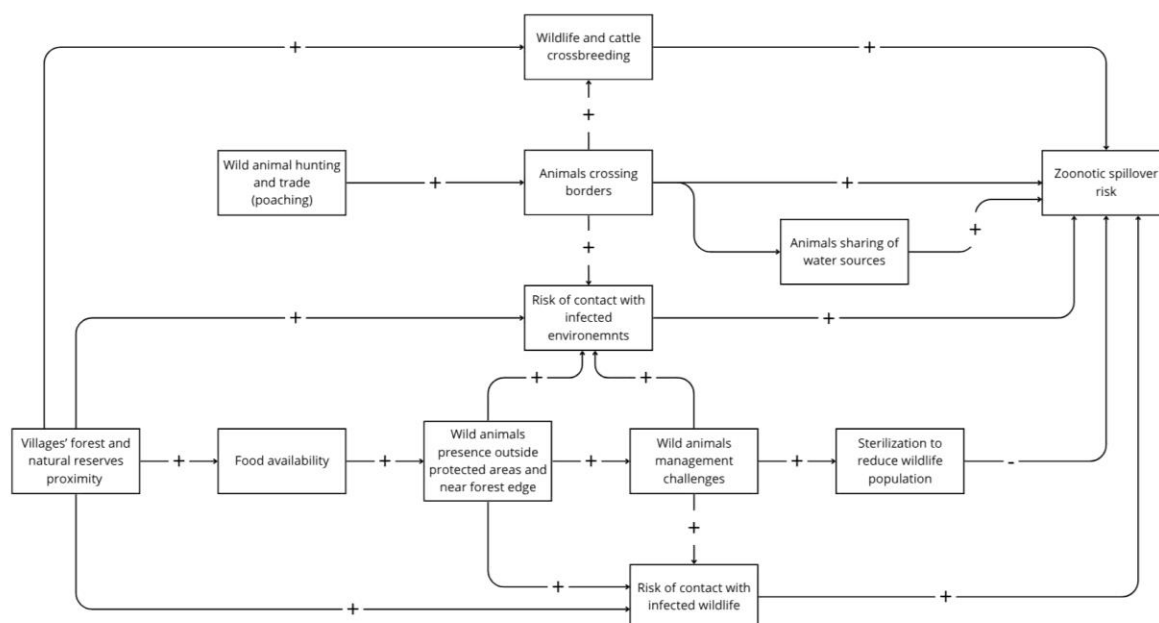


Figure 4. Relationship diagram 1: Animals and land border proximity

Table 1. Relationships and supporting quotes for the animals and land border proximity diagram

Relationships	Quotes from participants in FGD
Border proximity → Easy border crossing → Animals' movement → Disease Introduction Risk (+)	"There is a risk from animals that may come from outside the country, including diseases from wild animals. (...) Regarding animal health, the main risk is movement, which can introduce diseases. (...) Being near the border, animals or people can cross over easily. There is no real boundary, just a mountain that can be crossed on foot." (FGD 2)
Forest proximity → Contact with infected environments (+)	"For example, tourists might enter forests and unknowingly come into contact with infected areas, such as trees with hollows." (FGD 2)
Wild animals' presence outside protected areas and near forest edge → Manage wild animals outside protected areas → Sterilization to reduce wildlife population	"(...) policy of the Department of National Parks to manage wild animals outside protected areas. Initially, it was only elephants, but now monkeys are also being managed. There are measures such as sterilization to reduce the population." (FGD 1)
Border proximity → Wild animal trade → consumption of wild animal meat	"It's (bats) made into a dish called "Tom Phet." Mostly, it gets sent abroad to Vietnam." (FGD 1)
Communities' forest and parks proximity → Animals sharing of water sources → Red cattle and domestic cattle crossbreeding and Wild buffalo and domestic buffalo crossbreeding → Zoonotic spillover risk	"The issue is that pets live with people, and communities within the parks, such as wild buffalo and domesticated buffalo, may crossbreed. The main concern is the spread of epidemics (...). (...) The red cattle were once considered extinct, but now they are trying to increase the population. There are over 20 of them now, and there is a possibility of crossbreeding with domestic cattle, which could introduce new diseases because they share the same water sources." (FGD 1)

Food availability → Wild animals' presence outside protected areas and near forest edge

"The species that reproduce will stay at the forest's edge because they have a full food supply." (FGD 1)

Topic 2: Environmental issues and changes

One major pathway begins with climate change, which was described by participants as already having a tangible impact on local communities. Changes in temperature and rainfall patterns contribute to more frequent and intense flooding and droughts, both of which were seen to negatively affect public and animal health. One respondent noted, "*Thailand is one of the top countries in the world that is already being affected by climate change... when there are climate changes, such as flooding or droughts, it makes it easier for epidemics to spread. (FGD 1)*" Flooding increases the spread of waterborne pathogens by contaminating the water sources used by humans and animals. Participants pointed to diseases such as leptospirosis that become more prevalent in such conditions. "*With large floods, diseases like leptospirosis or other waterborne diseases can emerge, making it worse. (FGD 1)*" This contaminated water can become a direct route of zoonotic spillover, where pathogens are transmitted between species due to shared exposure to unsafe environmental conditions [82].

Droughts, on the other hand, were reported to lead to water scarcity, prompting villagers to store water. This, in turn, created breeding grounds for mosquitoes, especially when water was left standing for long periods. One participant explained, "*The drought has led to water storage, which in turn has created breeding grounds for mosquitoes. (FGD 2)*" These conditions were seen to exacerbate the spread of mosquito-borne viruses. Moreover, despite attempts to mitigate the issue through vector control measures like abate sand, uptake was limited due to acceptability concerns: "*There were difficulties with adding abate sand, as the smell was unpleasant, and villagers were unwilling to allow it... resulting in a shortage of water. (FGD 2)*" This illustrates how environmental and behavioral factors interact to increase the risk of vector-borne zoonotic spillover.

Another key pathway begins with agriculture and plantations, which drive forest encroachment and deforestation. Participants noted that the expansion of cultivated land has significantly reduced forest cover, contributing to broader ecological imbalance. "*The forest area has decreased... from 70,000 rai last year... Now it's down to 31%... This is due to environmental changes, where the area of forest has decreased, disturbing the ecological balance. (FGD 1)*" Such land use changes bring wildlife, livestock, and human populations into closer contact, disrupting natural buffers and increasing opportunities for pathogen spillover. These observations align with existing scientific literature on how land-use change is a significant driver of emerging infectious diseases.

Forest encroachment often leads to conflict between local communities and conservation authorities or goals, especially when livelihoods depend on forest access. The encroachment is not always intentional or malicious but reflects the reality of competing land needs for subsistence and livelihood, and it contributes to further ecological imbalance. As one interviewee explained, "*Sometimes it's understood that originally, they were encroaching on*

the forest, but now we're not blaming anyone... we're just asking how we can get the elephants back into their natural habitats while the people living around it maintain the balance. (FGD 1)

The diagram also includes a mitigation pathway, showing how efforts to promote human-animal coexistence could reduce tensions between community and conservation goals and contribute to ecological stability. Several participants mentioned community-based approaches, including crop changes and debt restructuring, which aim to reduce conflict with wildlife while preserving livelihoods: *"They suggested... to suspend their debts. Then, they would change their crops within five years and participate in the project. (FGD 1)"* Such efforts reflect a willingness to adapt, even if the changes are challenging: *"It requires adjustments, adjusting crops, adjusting lifestyles, but that means it will affect their economy and markets because they have to start over from scratch. (FGD 1)"* These perspectives highlight both the potential and the limitations of coexistence strategies in practice. While promoting coexistence may reduce ecological disturbances in the long term, it also requires coordinated support, incentives, and time for adaptation.

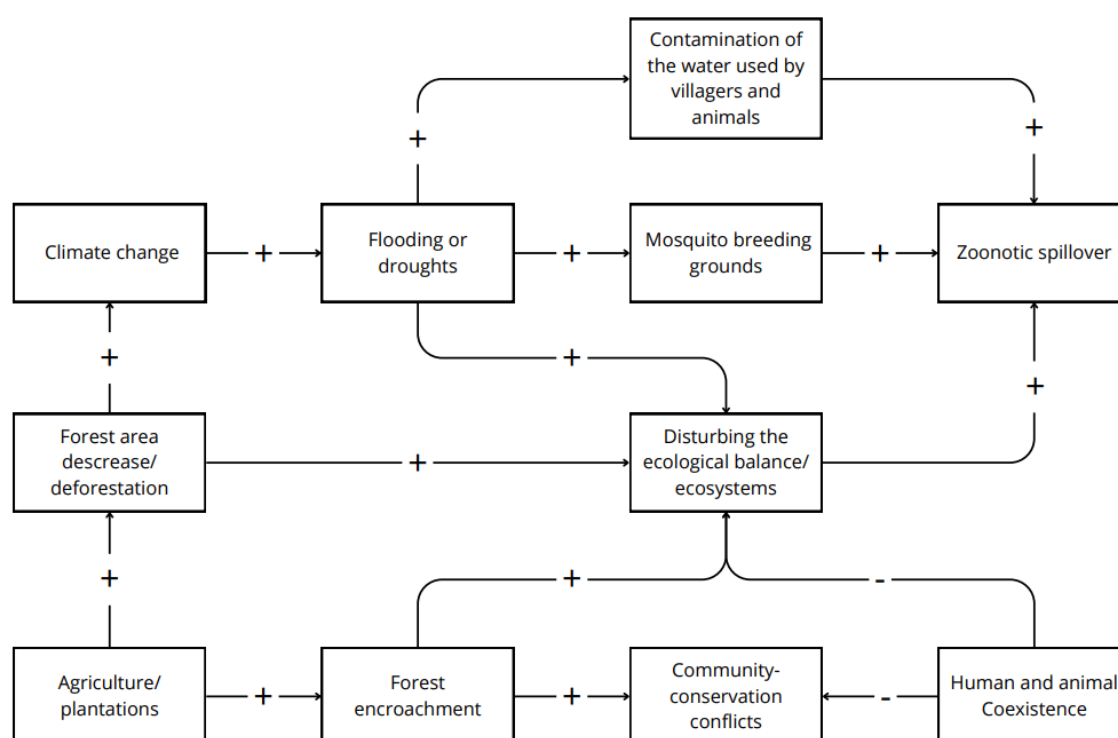


Figure 5. Relationship diagram 2: Environmental issues and changes

Table 2. Relationships and supporting quotes for the environmental issues and changes diagram

Relationships	Quotes from participants in FGD
climate change → flooding or droughts → contamination of the water used by villagers and animals → Zoonotic spillover (+)	"Thailand is one of the top countries in the world that is already being affected by climate change. For example, when there are climate changes, such as flooding or droughts, it makes it easier for epidemics to

	spread. Sometimes, with large floods, diseases like leptospirosis or other waterborne diseases can emerge, making it worse." (FGD 1)
flooding or droughts → mosquito breeding grounds → Zoonotic spillover (+) Measure: Adding abate sand	"This may be due to water levels or environmental factors that are suitable for mosquito breeding. If a mosquito carries the virus, it can lay infected eggs, and the newly hatched mosquitoes will also carry the virus, making it very difficult to control. Due to our environment and vector characteristics, the risk level here is higher than in some other areas. The drought has led to water storage, which in turn has created breeding grounds for mosquitoes. Efforts were made to address the issue, but there were difficulties with adding abate sand, as the smell was unpleasant, and villagers were unwilling to allow it to be poured out, resulting in a shortage of water. So, this has been an issue." (FGD 2)
Agriculture/ plantations → forest area decrease → disturbing the ecological balance → Zoonotic spillover (+)	"It may be that the environment in the area has changed. For example, in our country, the forest area has decreased like this from 70,000 rai last year. Now it's down to 31%. We still don't know what percentage of that is rubber plantation, or how much is actually forest, but they've mapped it out like that, so it's okay. This is due to environmental changes, where the area of forest has decreased, disturbing the ecological balance. This is one of the factors that may lead to emerging diseases." (FGD 1)
Agriculture/ plantations → Forest encroachment → Community-conservation conflicts (+)	"We just believe that everyone's life is equal. Now, how do we balance conservation with people living in their own way? We use the term "coexistence," but the meaning of coexistence varies for each person. Sometimes it's understood that originally, they were encroaching on the forest, but now we're not blaming anyone for that. We're just asking how we can get the elephants back into their natural habitats while the people living around it maintain the balance and don't go beyond that. This is part of it, and another part says, "Just take them out; I want my land." So, right now, it's about these two perspectives. Is there a perfect solution yet? No, there isn't." (FGD 1)
Human and animal Coexistence → Conservation tension → Disturbing the ecological balance/ ecosystems (-) Measure: crop adjustments/ changes	"They suggested asking the Bank for Agriculture and Agricultural Cooperatives (BAAC) to suspend their debts. Then, they would change their crops within five years and participate in the project. They proposed this and believed they could live with the elephants because, no matter what, the elephants are staying, and they feel safer near the forest's edge." (FGD 1) "At the same time, it requires adjustments, adjusting crops, adjusting lifestyles, but that means it will affect their economy and markets because they have to start over from scratch." (FGD 1)

Topic 3: Movement of people

The core driver in this system is labor mobility, particularly within the agricultural and construction sectors [83]. Seasonal workers, especially those involved in fruit production and packing, frequently move between provinces and across borders. This mobility heightens the risk of carrying endemic diseases, especially when health checks are absent or insufficient. One participant noted: " Likewise, *at the regional and provincial levels, policies promoting labor*

movement, such as to support agricultural production, are also driven by economic priorities. This can lead to disease spread when workers move in, and disease follows. (FGD 2) This can become worse with illegal migrant workers: *"There are issues with foreign labor... Sometimes, there are undocumented migrant workers who haven't gone through health checks, which increases the risk of bringing contagious diseases into our area. (FGD 2)"* This movement increases the risk of carrying endemic diseases, which may then circulate in new regions. A participant described the link between mobility and disease introduction as follows: *"When people move from one place to another, whether from one province to another or even from one village to another, they carry diseases with them. (FGD 1)"* In some cases, this movement has transformed disease patterns: *"Previously... Chanthaburi didn't really have Chikungunya... But when these workers moved south... they brought Chikungunya back with them. Since then... it has become a local issue. (FGD 2)"*

Pet abandonment is another consequence of mobile labor, particularly when temporary or seasonal workers leave their animals behind when relocating. *"Many workers travel back and forth... some keep pets like dogs and cats. When they leave, they often leave the pets behind, adding to the number of disease carriers in the area. (FGD 2)"* Abandoned pets, often unvaccinated or inadequately cared for, can become reservoirs or vectors of zoonotic diseases, further increasing outbreak risk through uncontrolled transmission among animal and human populations.

Another prominent factor captured in the diagram is illegal migration, which often intersects with public health risks in complex ways. Because undocumented individuals fear arrest and deportation, they may avoid seeking medical care, even when symptomatic. This leads to disease concealment, delays in diagnosis, and disease outbreak risks. One stakeholder reflected: *"In the case of migrants... they may have fled or entered illegally... they will hide [illness]... They were afraid to seek treatment. They were hesitant to provide full information. This shows... we may need alternative solutions to encourage them to seek services and treatment. Otherwise, it could lead to the spread of the disease. (FGD 1)"* This concealment of illness not only endangers the health of the individual but also increases the risk of broader community transmission. For example, one respondent described a delayed case of elephantiasis and another involving malaria in Bangkok, both of which were exacerbated by patients' reluctance to engage with the health system due to fear of legal consequences. These cases underscore the public health implications of criminalizing undocumented migration and suggest that a protective, rights-based approach to health access for migrants could reduce outbreak risks.

Tourism was also identified as a contributor to disease spread, particularly where incoming visitors are not subject to health screening or monitoring [84]. Tourism was similarly identified as a growing contributor to disease risk. While it is widely regarded as economically beneficial, participants highlighted concerns around limited health screening and oversight for incoming travelers. *"This poses a risk, along with people traveling from other countries who may bring various pathogens with them, including both tourists and laborers entering Thailand. (FGD 2)"* These concerns have intensified in response to recent policy changes: *"For example,*

with a visa-free policy, we see a significant increase in tourism. But the question is, with so many tourists coming in, we don't know what they may bring in with them (...). (FGD 2)" The lack of integrated public health measures in tourism planning was noted: "The focus tends to be on the revenue tourists generate (...) while health screening for diseases is rarely mentioned. (FGD 2)" As shown in the diagram, this increase in tourism feeds into the risk of carrying endemic or novel diseases, which, in the absence of detection and early response, raises the likelihood of local outbreaks. Such gaps in preventive infrastructure highlight the challenges of managing disease risk in regions with high levels of both formal and informal human movement.

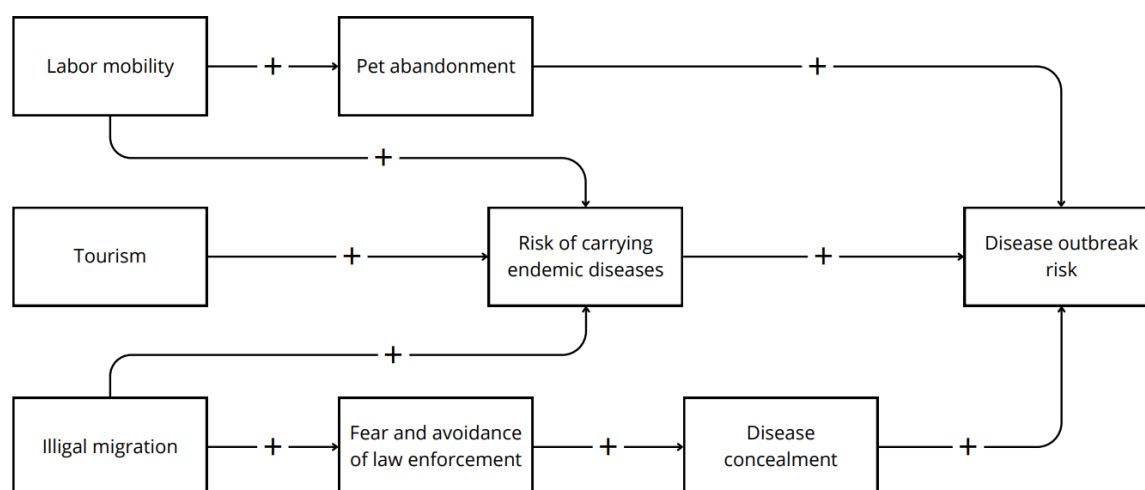


Figure 6. Relationship diagram 3: Movement of people

Table 3. Relationships and supporting quotes for the movement of people diagram

Relationships	Quotes from participants in FGD
Risk of carrying endemic diseases → Disease outbreak risk	"When people move from one place to another, whether from one province to another or even from one village to another, they carry diseases with them." (FGD 1) "Let's consider the host factor, specifically population movement, which has a significant impact. Monkeypox, for instance, is a difficult disease to control because we can't regulate who enters the province, including foreigners who come in without screening. This is a major factor." (FGD 2)
Tourism → Risk of carrying endemic diseases	"This poses a risk, along with people traveling from other countries who may bring various pathogens with them, including both tourists and laborers entering Thailand." (FGD 2) "For example, with a visa-free policy, we see a significant increase in tourism. But the question is, with so many tourists coming in, we don't know what they may bring in with them (...)." (FGD 2) "The focus tends to be on the revenue tourists generate (...), while health screening for diseases is rarely mentioned." (FGD 2)
Illegal migration → Fear of law enforcement → Risk of	"In the case of migrants, sometimes they may have fled or entered illegally, and the chances of them having a disease are high, but they will hide it. For example,

carrying endemic diseases → Disease concealment → Disease outbreak risk	if they have tuberculosis, they might claim they're seeking treatment for the costs or avoid contact with the police. In such cases, we coordinate to prevent the police from arresting them, but when they come in this way, it's not just about enforcing the law. The mechanism should be protective, helping them integrate into the system and making it safer for them to access services. In the past, some people hid their illness and let it get worse. One case involved someone with elephantiasis, and it took a long time to detect the disease because the person knew that if they were diagnosed with it, they would be sent back to their country. Their family members would also try to cooperate. When it comes to the law enforcement, it has consequences for those who enter illegally, but the challenge is how to make them feel confident enough to seek help or treatment. For instance, recently, there was a case where malaria was found in Bangkok, and the person was concerned about not having proper documentation and was afraid to seek treatment. They were hesitant to provide full information. This shows that while the law needs to be enforced, for vulnerable groups, we may need alternative solutions to encourage them to seek services and treatment. Otherwise, it could lead to the spread of the disease." (FGD 1)
Labor mobility → illegal migration	"Regarding migrant workers, I'd like to add something here. Since our province is known for fruit production, especially durian, there's a significant movement of laborers working in the fruit industry. For instance, once the durian season in the eastern region ends, they move south. Previously, if we're being honest, Chanthaburi didn't really have Chikungunya—there were very few reports. But when these workers moved south to places like Chumphon and Nakhon Ratchasima for durian work, they brought Chikungunya back with them. Since then, as mentioned, it has become a local issue. There's a lot of labor movement within this production cycle, in the packing houses, and throughout the fruit industry supply chain." (FGD 2) "Likewise, at the regional and provincial levels, policies promoting labor movement, such as to support agricultural production, are also driven by economic priorities. This can lead to disease spread when workers move in, and disease follows." (FGD 2) "There are issues with foreign labor. Sometimes, there are undocumented migrant workers who haven't gone through health checks, which increases the risk of bringing contagious diseases into our area." (FGD 2)
Labor mobility → pet abandonment	"There is also the issue of labor, which is similar to Chanthaburi, especially at the Mae Sai border. Many workers travel back and forth; some commute daily, while others come temporarily for construction work and even keep pets like dogs and cats. When they leave, they often leave the pets behind, adding to the number of disease carriers in the area." (FGD 2)

Topic 4: Knowledge and beliefs

At the core of the model is a health education gap, which was described as a fundamental barrier to effective disease prevention. This gap stems from both insufficient health education in schools and limited outreach within rural and ethnic communities. One participant explained: *"Another issue is public knowledge: people are often not well-informed*

about preventive measures. We hold occasional meetings, but preventive knowledge is still lacking. (FGD 2)”The need for early education was also emphasized: *“It needs to be integrated into the education system, so children grow up with this awareness... For example, in countries like Japan, they teach children how to respond to earthquakes... It would be good if they also included guidance on how to act during crises or emergencies. (FGD 1)”*The model shows that both a lack of structured health education and community-specific disparities contribute to a weak pandemic preventive culture, which underpins many of the risk pathways illustrated.

Linked to this is the influence of traditional medical practices, rituals, and beliefs, which further weaken positive attitudes toward pandemic preventive practices. These beliefs, while culturally embedded, can sometimes conflict with medical advice. For example: *“Some do not accept measures to prevent leptospirosis because of their traditional beliefs... For example, after being bitten by a dog, they might just wash the wound with water or treat it by hitting it with a horseshoe. There’s also a belief that... if a dog bites you, you should eat dog meat to avoid getting sick. (FGD 2)”* Such beliefs contribute to the normalization of high-risk behaviors and undermine community-wide disease prevention efforts.

This weak preventive environment leads to a series of specific behaviors that increase zoonotic risk: Raising wild and stray animals in household settings can introduce unknown pathogens and vectors into the community. *“Some households raise monkeys, or they take in dogs of unknown origin, leading to an increase in rabies cases. (FGD 2)”* Consumption of raw food was commonly cited as a culturally rooted but high-risk behavior. *“Some people consume these foods raw, and individual beliefs play a part. (FGD 2)”* Poor food preparation and storage practices were also highlighted as contributors to gastrointestinal and foodborne infections. A participant compared local practices with safer international models: *“In some foreign countries, oysters are passed through controlled water systems... In our country, many entrepreneurs lack the resources and knowledge... This is also an important dimension of disease transmission. (FGD 2)”*

Consumption of wild animals continues in some communities as a cultural norm or out of economic necessity. *“Birds, bats, and rodents... Some will get stuck [in nets]... so they cook it, fry it, grill it... This happened out of a lack of knowledge and awareness. (FGD 2)”*Eating fruits bitten by wild animals was another routine behavior observed in the absence of food safety guidance. *“There’s no inspection... so no one checks for contamination. They just cut off the bitten part. (FGD 2)”*

Occupational exposure among ethnic groups working in agriculture also increases zoonotic exposure. *“Jobs like rice farming increase vulnerability to leptospirosis or scrub typhus... almost all farmers work barefoot... changing behavior is challenging, which increases their risk of infection. (FGD 2)”* Finally, lack of animal vaccination, even for domestic pets, remains a critical and often overlooked issue. *“It may actually relate to a lack of public awareness. For instance, just getting their pets vaccinated—some people aren’t even aware they should. (FGD 2)”*Together, these behavioral and cultural risk factors form multiple reinforcing pathways that lead to elevated zoonotic spillover risk, as depicted in the diagram. At the center of these

relationships is the broader issue of insufficient health education, which underlies the persistence of high-risk practices across generations and communities.

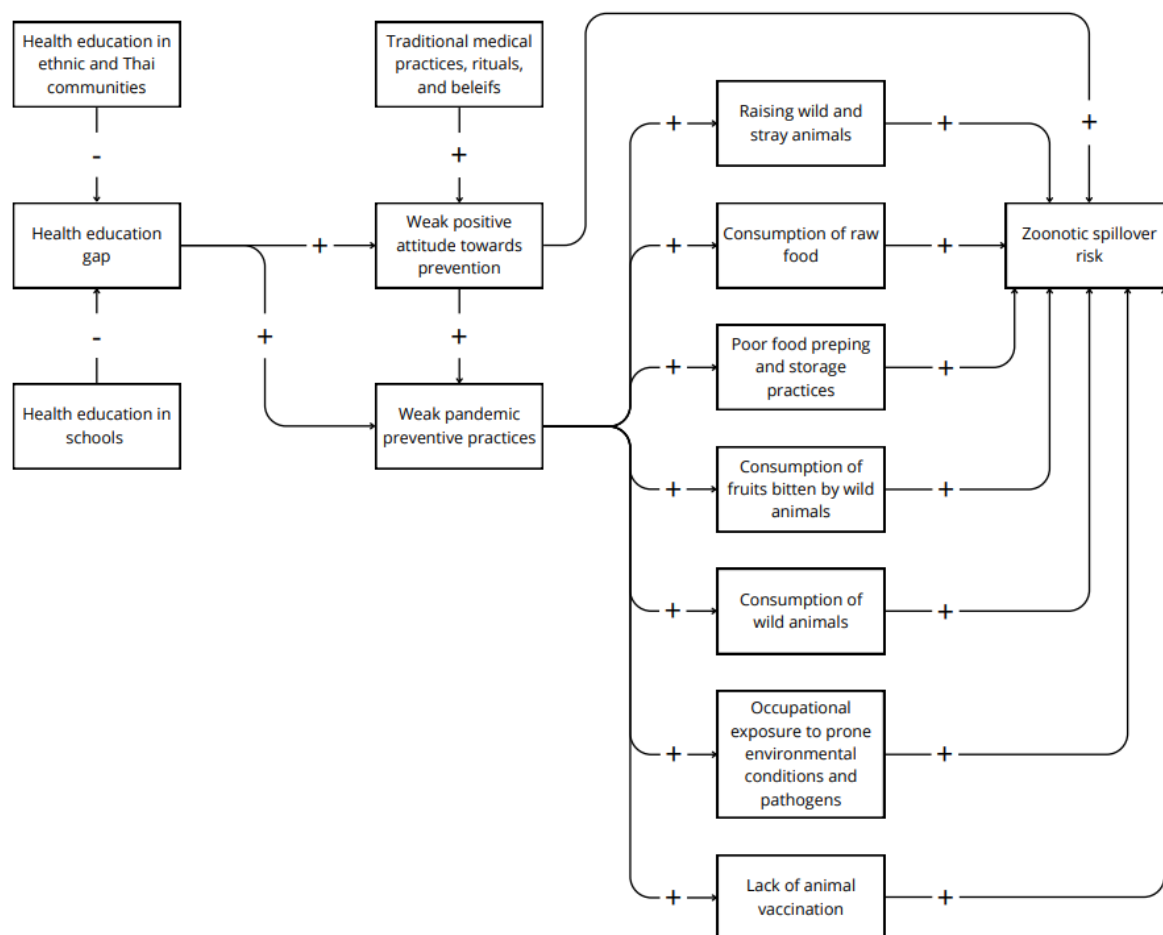


Figure 7. Relationship diagram 4: Knowledge and beliefs

Table 4. Relationships and supporting quotes for the knowledge and beliefs diagram

Relationships	Quotes from participants in FGD
Health education in ethnic and Thai communities → Health education gap → weak pandemic preventive practices	"Another issue is public knowledge: people are often not well-informed about preventive measures. We hold occasional meetings, but preventive knowledge is still lacking. These two factors are key issues." (FGD 2)
Health education in schools → Health education gap → weak pandemic preventive practices	"It needs to be integrated into the education system so children grow up with this awareness, starting from their generation. For example, in countries like Japan, they teach children how to respond to earthquakes through reviews and various forms of training." (FGD 1) "It would be good if they also included guidance on how to act during crises or emergencies, making it a part of daily life and ingrained in people's habits." (FGD 1)

Animal vaccination → Zoonotic spillover risk	"It may actually relate to a lack of public awareness. For instance, just getting their pets vaccinated—some people aren't even aware they should." (FGD 2)
Occupational exposure to prone environmental conditions and pathogens → Zoonotic spillover risk	"Occupational risk also plays a role. For instance, within ethnic communities, jobs like rice farming increase vulnerability to leptospirosis or scrub typhus. Protective practices, such as wearing boots, are uncommon—almost all farmers work barefoot. Environmental factors also add to these challenges. We even try to educate people about wearing boots while farming, but some say they've never worn them and don't want to start now. So, changing behavior in these ways is challenging, which increases their risk of infection." (FGD 2)
Consumption of wild animals and raw food → Zoonotic spillover risk	"Sometimes people sell wild animals and food products that may contain pathogens. People buy these wild animals, especially forest foods, from other areas, such as at markets where vendors from different villages sell them. Some people consume these foods raw, and individual beliefs play a part." (FGD 2)
Consumption of wild animals → Zoonotic spillover risk	"The main pests: bats, birds, these are the main enemies of farmers (...). When they put them (nets) up. Birds, bats, and rodents, right? Some will get stuck... so they cook it, fry it, grill it, stir-fry it, or make curry. It's according to the local menu. This happened out of a lack of knowledge and awareness." (FGD 2)
Consumption of fruits bitten by wild animals → Zoonotic spillover risk	"Yes, and there's no inspection (referring to fruit trees attacked by bats), so no one checks for contamination. They just cut off the bitten part. If the damage is extensive, they throw it away." (FGD 2)
Poor food prepping and storage → Zoonotic spillover risk	"In addition, there are cases of foodborne illnesses and gastrointestinal infections. For instance, during an investigation in Phang Nga, <i>Vibrio parahaemolyticus</i> was found. The public health department realizes that contracting these diseases can be linked to eating habits, such as consuming raw food. However, if we look deeper, it raises questions: why can people eat raw oysters abroad without getting food poisoning? This might relate to differences in agricultural and farming practices. For example, in some foreign countries, oysters are passed through controlled water systems before being sold, helping to maintain cleanliness and manage temperature, thereby reducing direct human exposure to pathogens. In our country, however, many entrepreneurs lack the resources and knowledge to implement such practices. If these factors were addressed, it could help economically by providing high-quality food

	ingredients while preserving the delicacy of local cuisine, allowing people to eat raw foods safely. Therefore, I think this is also an important dimension of disease transmission." (FGD 2)
Traditional medical practices, rituals, and beliefs → Weak positive attitude towards prevention → Zoonotic spillover risk	"Yes, there are issues related to traditional beliefs and customs in some villages within this community. For example, some do not accept measures to prevent leptospirosis because of their traditional beliefs passed down through generations, including a perception that injections are dangerous." (FGD 2) "Sometimes, people believe things that the doctor said were misconceptions. For example, after being bitten by a dog, they might just wash the wound with water or try to treat the wound by hitting it with a horseshoe, believing that it will prevent illness. There's also a belief that if a dog bites you, you should eat dog meat to avoid getting sick. These are superstitions that persist." (FGD 2)
Raising wild and stray animals → Zoonotic spillover risk	"As villagers sometimes raise wild animals, which could lead to future issues. Some households raise monkeys, or they take in dogs of unknown origin, leading to an increase in rabies cases" (FGD 2)

Topic 5: Use of information technology

A central theme in the diagram is the use of information technology to improve the ability to adapt to isolation due to sickness, which was highlighted as a critical issue during the COVID-19 pandemic. Stakeholders emphasized that technology allowed people to continue working, learning, and accessing services despite movement restrictions. *"COVID-19 has shown us how many obstacles exist—people couldn't leave their homes... However, technology helped sustain livelihoods... like traveling to Bangkok virtually. This highlights how technology could serve as a key tool for planning and preparedness in the future. (FGD 1)"* By supporting remote interaction and continuity of services, IT enhances individuals' resilience during crises, which contributes positively to future pandemic preparedness.

IT also plays a critical role in enabling effective health communication, which is essential for reducing misinformation and building public trust during crises [85], [86]. Participants emphasized that digital platforms allowed for timely alerts that shaped public perception and helped prevent panic. For example, one respondent recalled: *"The latest incident at Paragon Department Store, where they talked about sending warning SMS messages, they actually did. (FGD 1)"* In another case, media framing was used strategically to ease tensions during a sensitive situation: *"The media presented it in a positive light, and it turned out that it helped reduce the tension... Even though elephants were stomping on and killing cows... (FGD 1)"* These examples underscore how well-coordinated communication, supported by digital tools, can mitigate anxiety and maintain social cohesion during emergencies.

At the same time, participants also pointed to limitations in communication infrastructure, particularly during high-stress periods like the COVID-19 pandemic. *“It made us realize that everyone would flood the hotlines with calls... even with the hotline and other resources, it turned into a chaotic situation... (FGD 1)”* This illustrates that digital communication systems need to be designed for scalability and clarity, or they risk becoming bottlenecks during critical moments. Moreover, stakeholders raised concerns about the role of social media in fueling misinformation and social division. As one explained: *“Sometimes we respond based on our personal opinions, but it can turn into a trend in society... media that promotes drama—like the stories that go viral—turns the situation into a political issue. (FGD 1)”* This highlights the dual nature of information technology: while it has the power to disseminate accurate information quickly, it also carries the risk of amplifying confusion or conflict if not used responsibly.

On a systems level, information technology strengthens data sharing between institutions, a critical enabler of collaboration, resource efficiency, and real-time response. *“In the past, there was no sharing of data, and people collected data separately... This means the department doesn’t need to spend its budget on it. (FGD 1)”* When institutional data is accessible and aligned, it reduces duplication of effort, which in turn improves budget and resource management. Efficient use of resources supports better outbreak management, ultimately feeding into stronger preparedness for future health crises. In contrast, when data sharing is fragmented, institutional coordination breaks down, delaying local response and decision-making: *“Letters are sent to the provincial governor... But the local units might not even know it exists. By the time a notice or warning reaches the public, the outbreak could be over. (FGD 2)”* This bottleneck in communication underscores the importance of IT systems that are not only centralized but also responsive across vertical administrative structures.

In terms of epidemiological surveillance, the use of technology supports improved outbreak mapping, allowing for more targeted, evidence-based decisions. *“There’s a map indicating where outbreaks are happening... Before we move anywhere, we need to know if there’s any risk... This is how we manage it. (FGD 1)”* These tools enhance effective decision-making, which strengthens outbreak response and overall system readiness. Whether for local travel guidance or cross-sector coordination, technology enables timelier data-driven action. The diagram, therefore, captures multiple reinforcing relationships, where the use of IT leads to improvements in isolation adaptation, communication, data coordination, and decision-making, all contributing to a stronger, more agile public health system. Each of these pathways ultimately feeds into the goal of future pandemic preparedness.

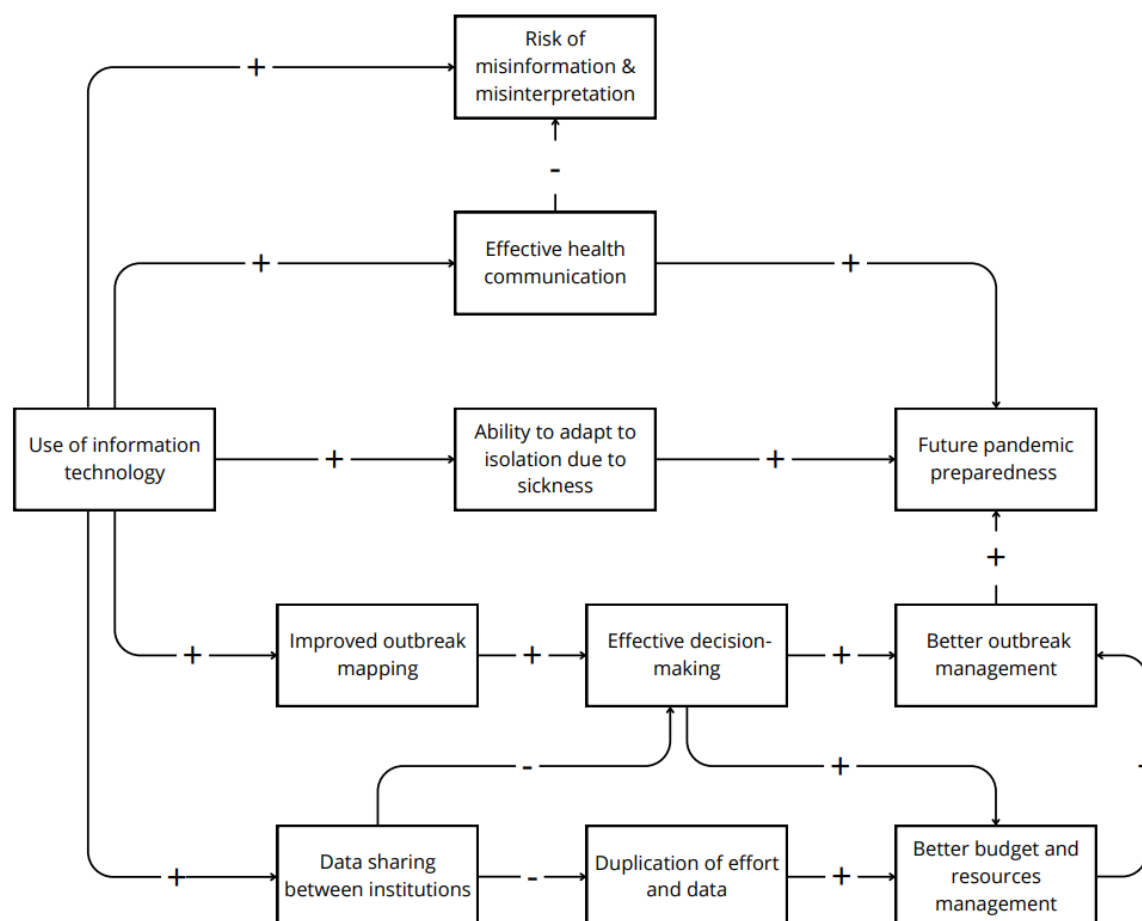


Figure 8. Relationship diagram 5: Use of information technology

Table 5. Relationships and support quotes for the use of information technology diagram

Relationships	Quotes from participants in FGD
Use of information technology → Ability to adapt to isolation due to sickness → Future pandemic preparedness	"From a technology perspective, COVID-19 has shown us how many obstacles exist—people couldn't leave their homes or make a living. However, technology helped sustain livelihoods and allowed life to continue in unexpected ways. For instance, we found ways to work remotely and adapt to new realities, like traveling to Bangkok virtually. I think this highlights how technology could serve as a key tool for planning and preparedness in the future. It eventually becomes the "normal," incorporating the use of various technologies." (FGD 1)
Use of information technology → Data sharing between institutions → Duplication of effort and data → Better budget and resources management → Better outbreak management → Future pandemic preparedness	"We often have to ask again and collect new data. But in reality, should data or Big Data be managed at the local level or by the organization, and who should be at the center? (...) So, there's also the issue of sharing. (...) And it is a process of collecting duplicate data and spending the budget on the same issue. In the past, there was no sharing of data, and people collected data separately. But in our case, the way we use data is that we collect the data and then share it with the department. We invest in gathering data, such as elephant data, and then share it with the department. This means the department doesn't need to spend its budget on it. The budget that we couldn't secure is saved, and I think there should be more data sharing at some level, which would enable

	analysis. When the analysis is based on the same level of data, it can be more effective." (FGD 1)
Use of information technology → Data sharing between institutions → Effective decision-making → Better budget and resources management → Better outbreak management → Future pandemic preparedness	"Take, for example, projects like avian flu or rabies. Letters are sent to the provincial governor, requesting collaboration across three or four agencies. But coordination problems arise because, while a letter reaches the top, the local units might not even know it exists. This is one real issue on the ground. Sometimes, when it finally reaches the local level, they're unsure if it's even part of their duties." (FGD 2) "It becomes a chain of individual units holding responsibility until it reaches the ministry level, then across ministries, which involves additional regulatory review. This whole process takes an enormous amount of time because of all the document handling and format requirements. For instance, by the time an official notice or warning about a disease reaches the public, the outbreak could be over. Even with "urgent" stamped on it, the entire process is far from being truly urgent." (FGD 2)
Use of information technology → Improved outbreak mapping → Effective decision-making → Better outbreak management → Future pandemic preparedness	"There's a map indicating the locations where outbreaks are happening. Therefore, there is a risk management process where we need to check beforehand if it's safe to go. We need to constantly collect data, and we also have to keep monitoring. Before we move anywhere, we need to know if there's any risk of contracting a disease in that area. This is how we manage it." (FGD 1)
Use of information technology → Effective health communication → Future pandemic preparedness	"The latest incident at Paragon Department Store, where they talked about sending warning SMS messages, they actually did." (FGD 1) "The veterinary hospital came up with a method of photographing elephants when they go into the market in a way that makes them look cute. The media presented it in a positive light, and it turned out that it helped reduce the tension in Kaeng Krachan. Even though elephants were stomping on and killing cows, there was no news coverage because the images portrayed it positively." (FGD 1) "It (COVID-19 pandemic) made us realize that everyone would flood the hotlines with calls, making it impossible to provide service effectively. The hotline, which was supposed to gather and relay information, initially seemed like a good idea. But in the end, it highlighted issues in communication systems and the lack of a seamless flow. The idea was for information to flow efficiently through the system, but even with the hotline and other resources, it turned into a chaotic situation where everyone was overwhelmed, competing for attention, and eventually burning out. People couldn't access what they needed." (FGD 1)
Use of information technology → Effective health communication → Risk of misinformation & misunderstanding	"On social media, as users, sometimes we respond based on our personal opinions, but it can turn into a trend in society. I think we need to build resilience in how we use media in society to improve understanding of the issue. If people understand the problem correctly, there will be fewer arguments and no drama. The issue can be solved, but when drama takes over, it becomes harder to resolve. However, media that promotes drama—like the stories that go viral—turns the situation into a political issue," (FGD 1)

Potential Intervention Matrix

The identified potential interventions are organized into five key areas. Each table summarizes stakeholder-identified problems, suggested measures, implementation pathways, and expected impacts. These findings are grounded in participant perspectives and supported by illustrative quotes included alongside each entry.

Area 1: Potential interventions at the community level

Table 6 suggests community-centered potential interventions designed to address interconnected health, economic, and environmental challenges. Several measures focus on empowering individuals and communities through targeted financial support and participatory approaches. For example, the development of insurance schemes by government agencies and private partners aims to protect local communities from the financial risks of property damage and loss of life [87]. Similarly, welfare programs and direct subsidies are proposed to assist individuals unable to work due to illness, thereby promoting economic stability and reducing hardship for vulnerable families.

Other potential interventions could leverage the collaboration with trusted community institutions, such as temples, in both financial aid distribution and public health communication. Temples may serve as effective platforms for reaching marginalized groups, especially the elderly, and facilitating support during crises [88]. Campaigns to promote lifestyle adjustments in response to zoonotic risks also emphasize the need for community engagement [89]. Altogether, these strategies reinforce the value of inclusive, locally grounded interventions that build resilience and foster trust within the community.

Table 6. Potential interventions at the community level

#	Quote	Problem Identified	Measure/ Intervention Suggested	How (Who & How)	Impact (What & Who)
1	"In terms of solving the problem, who will help? Who will compensate for the damage to property, the loss of life? ... This is why they are trying to consider insurance development as an innovation to assist" (FGD 1)	Lack of financial protection for property and life	Develop insurance schemes for loss compensation	Government agencies partner with insurance companies to develop schemes for communities	Financial security and reduced economic vulnerability for local communities

2	"At the same time, it requires adjustments—adjusting crops, adjusting lifestyles—but that means it will affect their economy and markets because they have to start over from scratch. Starting over is difficult, and we're not sure if they will agree with us." (FGD 1) "They suggested asking the Bank for Agriculture and Agricultural Cooperatives (BAAC) to suspend their debts. Then, they would change their crops within five years and participate in the project. They proposed this and believed they could live with the elephants because, no matter what, the elephants are staying, and they feel safer near the forest's edge." (FGD 1)	Community practices increasing zoonotic spillover risks	Campaigns promote lifestyle adjustments and new practices. The campaigns need to offer farmers something in return, recognizing that such big changes have a large economic and lifestyle impact. For instance, debt suspension in the case of crop changes.	Partnership between public health officials from the subdistrict health promoting hospital (SHPH), community leaders, and banks (Bank for Agriculture and Agricultural Cooperatives).	Reduced community and individual risk behaviors, a decrease in human-animal conflict, and long-term resilience.
3	"Another issue is that there is still a gap in supporting people who are locked down and need to stay at home. Some of them earn a daily income and can't go anywhere. I've seen how they've solved the problem through the public health system; they reached out to temples. Many people donated, and it became like the monks gave, and they used the donations for a while. They also spread the word, asking others to help. When we helped them, we provided donations as well. It's still lacking though; there's still a need for support in this area." (FGD 1)	Economic hardship due to sickness	Provide direct financial support to individuals who lose income due to sickness.	Government welfare programs distribute aid or subsidies to individuals who can't work due to sickness, which can be supported by a doctor's note confirming the need for sick leave.	Improved economic stability, reduced hardship for affected families
4	"For example, in the case of monks, when they have the capability to handle situations and can reach communities or elderly people, it all ends up being blocked at the point of getting the funding to support the monks' work or to purchase resources or bring in service units. So, it still dies at that point. Government funds can't be used for temples." (FGD 1)	Government funds can't be used for temples	Engage temples in health communication and education initiatives	Government collaborates with temples to reach communities and especially the elders	Increased trust and cooperation from the community

Area 2: Potential interventions covering animal interaction

Table 7 outlines a range of animal-focused potential interventions aimed at mitigating zoonotic disease risks and improving disease control systems. Several measures target

regulatory improvements, such as stricter law enforcement against high-risk wildlife carriers and enhanced pet ownership regulations. These potential interventions aim to reduce the likelihood of zoonotic spillovers and improve public safety, with local governments, law enforcement, and legislative bodies playing key implementation roles. For instance, introducing pet registration regulations and enforcing penalties for irresponsible ownership [90] would lead to better control of animal populations and reduced disease transmission risks.

Furthermore, stakeholders emphasize the importance of institutional coordination and improved logistics. For example, addressing fragmented communication between wildlife-related agencies through integrated platforms could strengthen surveillance and response capacity. One of these agencies mentioned by stakeholders is the Department of National Parks, which is responsible for managing wild animals outside protected areas (e.g., elephants, monkeys) using sterilization to control populations. Another measure proposed by stakeholders is the establishment of national vaccine management guidelines for pets, which would ensure more consistent and timely rabies vaccination [91] across localities. Collectively, these ideas for interventions reflect the need for stronger governance and collaboration across sectors to manage animal health more effectively and prevent public health threats.

Table 7. Potential interventions covering animal interaction

#	Quote	Problem Identified	Measure/ Intervention Suggested	How (Who & How)	Impact (What & Who)
1	"People who illegally remove wild animals and change the environment in any way that could lead to an epidemic should be punished more severely." (FGD 1) "If we prioritize animals that are carriers of diseases from the wild, such as bats, it's a problem to collect bat droppings. Now, there might be a need for stricter enforcement on these wildlife carriers." (FGD 1)	Ineffective law enforcement against practices that increase disease risk and higher disease risk in certain animals, such as bats	Implement targeted measures related to animals that present a higher risk of zoonotic spillover: higher legal punishments or fines for activities involving these animals	Local government, judiciary, and police to enforce fines and penalties	Reduced zoonotic spillover risk due to reduced illegal activities and enhanced compliance by the local community
2	"The concern we have about the outbreak of diseases, such as the case with cattle, comes from natural channels where they are smuggled to avoid taxes. If we were to allow only one legitimate channel, we could control and monitor it. This may require tax measures and possibly reducing or liberalizing imports, so we can control it once it crosses borders. It's widely known that these goods are being	Burdensome taxation affecting disease control efforts	Provide tax incentives or relief to entities complying with regulations	Ministry of Finance in collaboration with Health Departments implement targeted tax policies	Increased compliance, improved economic viability of health measures

	brought in through certain investment groups or areas." (FGD 1) "If we allow them to enter correctly with a single measure, such as reducing taxes or using lower taxes, they will be more willing to comply, and we will be able to monitor and control it." (FGD 1)				
3	"The communication of agencies related to wildlife should extend beyond the local area. There should be a special task force directly managing this issue, such as the Department of National Parks. This could be an opportunity, and we could also use more stringent laws. The effectiveness of this could be an opportunity. They might need a complete overhaul. Elevate it. A reform. This is one opportunity that could lead to legal reform as well." (FGD 1)	Fragmented communication between agencies working in the wildlife sector	Develop integrated communication channels between wildlife agencies	Government authorities facilitate regular communication platforms	Improved inter-agency cooperation and disease control effectiveness
4	"For instance, laws on dog ownership don't receive much attention. (...) But our laws don't regulate dog ownership as strictly as, say, in China, where owners must register their dogs. Here, anyone can own a dog, and stray dogs can roam, bite people, and cause chaos without repercussions." (FGD 2)	Ineffective pet ownership regulations	Regulate pet ownership with pet registrations	Government legislative bodies implement pet registration regulation	Reduced disease risk and better pet ownership management
5	"The rabies vaccine is administered by local administrative organizations (LAOs) only once a year. However, they should ideally have a stock of vaccines available whenever new puppies or kittens are born, which can happen every two to three months. Yet, local organizations only vaccinate once a year and do not keep a stock." (FGD 2)	Insufficient rabies vaccination and stock	Establish national vaccine management guidelines for pets, especially in diseases that can be transmitted from animals to humans	Ministry of Health develops clear guidelines and local authorities implement	Consistent and effective vaccine distribution

Area 3: Potential health-focused interventions

Table 8 presents a comprehensive set of health-related potential interventions aimed at strengthening public health systems at both the community and institutional levels. Understanding how Thailand's health system is organized is crucial to appreciating how these interventions can be effectively implemented. The Ministry of Public Health (MOPH) oversees national policies and legislation, such as the Public Cleanliness Act, and is responsible for the

curriculum of Village Health Volunteers (VHVs). Everything is operationalized through a network of provincial, district, and subdistrict structures. At the subdistrict level, we have the Subdistrict Health Promoting Hospital (SDHPH), which serves as a key local primary care unit. SHPHs are responsible for performing 14 Essential Public Health Functions (EPHFs), including health promotion, disease control and prevention, health assessment and primary medical care, rehabilitation, home visits, laboratory and pharmaceutical services, consumer protection, occupational health and environmental services, Thai traditional and alternative medicine, quality and academic development, primary health care, and public health law enforcement [92].

Stakeholders stress the need to improve the training and empowerment of VHVs through more hands-on, practical instruction and curriculum alignment with government goals, ensuring volunteers are prepared to respond effectively to local health challenges. VHVs provide primary healthcare to several households in a community, and their roles include correcting health misinformation, conducting surveillance on communicable and chronic diseases, offering first aid, promoting rational drug use, and supporting broader health education campaigns [93]. Their proximity to households allows them to monitor health conditions holistically and act as an early-warning system for public health threats. They also serve as liaisons between communities and SDHPHs, ensuring that health messages, services, and interventions reach even the most remote populations [73]. However, for these volunteers to be truly effective, they must be well-trained, supported, and incorporated into formal policy and planning processes.

The importance of governance and institutional support was also evident in the participant's statements. Potential interventions include stricter enforcement of existing public health laws, to reduce environmental and zoonotic disease risks. To support implementation, stakeholders emphasized the need for greater central support to local health agencies in the form of training, resources, and funding. Clear and shared responsibilities at the subdistrict level are also recommended, suggesting the development of a targeted disease control plan where agencies work together with assigned roles to improve coordination and accountability.

Lastly, stakeholders highlighted a critical gap in disease preparedness and surveillance. It calls for the allocation of dedicated national budgets and infrastructure to support proactive disease monitoring [94], rather than reacting only once outbreaks occur. Integrating health goals and indicators into 5-year and annual development plans is also proposed to ensure that public health remains a priority, guiding sustained and measurable action at the local level. Stakeholders identified Provincial Administrative Organizations as responsible for developing 5-year strategic plans. Together, these interventions aim to build a more resilient, responsive, and integrated health system. Another key recommendation is the integration of modern medical knowledge with traditional beliefs [95] to raise awareness and foster evidence-based resilience within communities. Thai traditional and alternative medicines are recognized as essential public health functions in Thailand's SDHPHs, with services such as herbal remedies and consultations encouraged by national policy, however, their implementation remains limited due to a shortage of qualified practitioners, reliance on minimally trained assistants, and

the unavailability of key services like herbal steam baths in many area [92]. In rural Guatemala, *casas maternas* provide a practical example of integrating modern maternity care with traditional beliefs by allowing *comadronas* to perform cultural birthing practices alongside trained medical staff in home-like birthing centers [96].

Table 8. Potential health-focused interventions

#	Quote	Problem Identified	Measure/ Intervention Suggested	How (Who & How)	Impact (What & Who)
1	“Looking at the integration of modern medical knowledge and traditional beliefs, there’s a need to adapt and prepare for any future outbreaks. Traditional beliefs may not have been scientifically proven, but they can be evaluated using modern medical knowledge. If this integration is achieved, it can raise awareness among communities, providing them with evidence-based information. This, I believe, is a way to build knowledge-based resilience.” (FGD 1)	Lack of integration of modern medical knowledge and traditional beliefs	Combine modern medicine with traditional beliefs as a form of preparedness for future outbreaks	Ministry of Health and SDHPH	Raised awareness among communities, providing them with evidence-based information, and building knowledge-based resilience.
2	“However, when it comes to the government sector, it often takes time to develop volunteers. Sometimes, we want volunteers to be ready to work immediately, so it feels like they learn better through hands-on experience than through the curriculum itself. At times, the curriculum and the volunteers don’t quite align. (...) We need to focus on empowerment alongside developing the curriculum, which must be aligned with the government to ensure sustainability moving forward.” (FGD 1)	Insufficient number of VHVs and insufficient training capacity: more hands-on training and curriculum development	Curriculum revision and development, inclusion of more practical training, and an increase in training capacity.	The curriculum is established by the Ministry of Public Health, and they are closely supervised by healthcare officers in SDHPHs. There could be a specific government branch dedicated to the training and development of VHVs	Improved public health response and increased prevention of disease outbreaks
3	“The health assembly in each province is not the same. Each province may have provincial assembly issues. If we can take these issues, gather a group of people, and really look at the local problems, there are many people who can organize the issues. Then, these	Lack of integration of local health issues at the provincial level, as local health	Facilitate structured community consultations and issue-gathering sessions to	Local health agencies and community organizations collaborate with the National Health	Community voices are better represented in provincial health planning,

	issues can be brought into the Provincial Assembly. The National Health Commission (NHC) provides some financial support, so it can be included in this." (FGD 2)	concerns may not always be captured or presented effectively across provincial health assemblies.	identify local health problems that can be formally submitted to the Provincial Health Assembly.	Commission (NHC) to organize inclusive local forums. These forums gather input, prioritize issues, and prepare submissions for the provincial assembly, using financial support provided by the NHC.	leading to an improved recognition of local needs, and health policy becomes more participatory and locally relevant.
4	"Sometimes, disease control responsibilities fall on the Subdistrict Administrative Organization (SAO), which may claim a lack of personnel, budget, or resources, shifting the responsibility to the Subdistrict Health Promotion Hospital or district public health officers, even though it should be their role. But if we were to conduct a SWOT analysis, bringing together all agencies to identify strengths, weaknesses, and areas for improvement, and develop a targeted disease control plan, it would be very beneficial. (...) An advantage would be having a shared plan where roles and responsibilities are clear. Each department would know their specific tasks to achieve a common outcome. This clarity strengthens coordination." (FGD 2) "Actually, if we work together with a unified goal, the benefits are already there. We share a clear direction, and each person knows their specific responsibilities and works simultaneously toward that goal. " (FGD 2)	Unclear disease control responsibilities at the subdistrict level.	Develop a targeted disease control plan, where roles and responsibilities are clear, so each department knows their specific tasks to achieve a common outcome.	Bringing together all subdistrict agencies to identify strengths, weaknesses, and areas for improvement in disease control and establish a plan where roles are clearly assigned	Strengthened coordination, increased clarity, accountability, and efficiency in disease management
5	"For example, they might assign certain tasks without truly incorporating health into the strategy. This makes it very difficult to integrate health goals into their 5-year strategy and action plan. Typically, they focus on annual plans: what they'll do, how, and to what extent. However, these plans lack a strong	Health is not prioritized or integrated into local development strategies and plans	Integrate clear health goals and indicators into plans.	Local government planning units work with health officials to embed health targets and indicators into 5-year and annual plans	Local health outcomes improve, plans become measurable and accountable health actions

	public health focus with specific health indicators, goals for disease reduction, or any measurable outcomes. As a result, if health is not a priority in their plan, there's no way to ensure it gets implemented." (FGD 2)				are more likely to be implemented
6	"Another thing I would like to see from the central level is a greater emphasis on disease surveillance and preparedness. Right now, budgets for setting up surveillance systems are minimal, with most funding focused on reacting only after an outbreak occurs. If we start from the beginning by prioritizing disease surveillance, it should help prevent outbreaks and reduce costs for later control measures." (FGD 2)	Under-funded, fragmented surveillance & preparedness capacity	Allocate a dedicated national budget for surveillance infrastructure & training	The central government prioritizes disease surveillance and preparedness, setting up a surveillance system and allocating funds	Early detection avoids large-scale outbreaks, saving future control costs for society

Area 4: Potential interventions related to policy making and governance

Table 9 highlights law and governance-related interventions aimed at improving coordination, policy design, and implementation across sectors and administrative levels. A central recommendation was to ensure that community members have a participatory role in the development of laws and policies. This includes aligning legislation with the lived realities of communities through consultation and collaboration among local governments, civil society, and relevant sectors such as health and veterinary services. When laws reflect local realities, communities become more engaged, and policies are more effective.

Another key focus is the need for clearer and more standardized operational protocols. When local stakeholders are unsure about what to do, especially during outbreaks, health authorities should provide clear guidance and training. This reduces confusion and ensures better preparedness. Additionally, the table points to the importance of regularly reviewing policies to avoid unintended consequences. Policymakers are encouraged to adapt policies proactively by assessing their impacts and addressing issues that arise when public health priorities are overshadowed by economic concerns.

Finally, the table calls for integrated and flexible policymaking that accounts for diverse local contexts. Rather than imposing rigid national policies, provincial and district-level authorities should be allowed to tailor responses based on local characteristics, with central governments offering guidance and oversight. Similarly, securing border areas to prevent illegal entry of people and animals is seen as a critical legal intervention to prevent cross-border disease spread [97]. Altogether, these measures aim to create a more coordinated, context-sensitive, and responsive policy environment that strengthens governance and supports long-term public and animal health outcomes.

Table 9. Potential interventions related to policy making and governance

#	Quote	Problem Identified	Measure/ Intervention Suggested	How (Who & How)	Impact (What & Who)
1	"Another important aspect is that we may need to listen to people more. For example, if we create laws without listening to people, we might miss understanding why they need to consume certain foods or hunt animals. They may have valid concerns that we need to address. Policymakers should open their minds to listen and learn from them, to understand why they act a certain way and to create policies that are aligned and integrated with their needs." (FGD 1) "In each province, there are certain organizations that can be involved. (...) speaking in terms of civil society language, there's "locality" and "community," but it seems there's no real local involvement. The village headman mentioned does not contribute to the proposal." (FGD 2) "There should indeed be a role for community members or residents in this process." (FGD 2)	Laws disconnected from the community's reality	Development of laws where experts and community members have a participatory role. Establish coordination and communication between government agencies and community members	Local governments coordinate meetings between the health, wildlife, veterinary sectors, and community members	Community has a participatory role, and community and government agencies become more connected
2	"Even if you want to issue centralized orders, the lack of clarity is a problem. You need to be bold and communicate clearly, specifying steps 1, 2, 4, 5 explicitly. Otherwise, unreliable information or uncertainties create questions that hinder effective action. To address this, the government or relevant authorities must reflect and adapt. You're not the one controlling the game; instead, you must bring everyone together because you are the communicator. (...) The goal should be to create a system that functions independently, unaffected by who moves, changes roles, or leaves. If that could be achieved, it would be ideal." (FGD 1)	Lack of clear and effective operational protocols	Develop and disseminate clear, standardized protocols	Health authorities develop clear protocols and train local stakeholders	Better preparedness, standardized actions during outbreaks

3	I think policy itself can be a risk factor. It's not just local, but also national policy. The focus tends to be on the revenue tourists generate, while health screening for diseases is rarely mentioned. (...) Likewise, at the regional and provincial levels, policies promoting labor movement, such as to support agricultural production, are also driven by economic priorities. This can lead to disease spread when workers move in, and disease follows." For example, with a free visa policy, we need to examine the health status of incoming individuals—will there be any health risks associated with them? The same goes for policies regarding the influx of labor. We need to be mindful of these issues. It's not just about responding to incidents after they happen and relying on public health measures, as public health tends to be reactive. (FGD 2)	Poorly designed policies inadvertently increasing risks	Regularly review and adjust policies to address unintended consequences	Policymakers conduct impact assessments and revisions	Reduced unintended consequences, improved policy effectiveness
4	"There isn't an integrated policy approach, which complicates problem-solving. While we have certain indicators to meet, the indicators from the relevant agencies don't always align." (FGD 2) "However, when it comes to diseases that can transfer from animals to humans, we can't work in isolation. We must coordinate with other entities. (...) There should be more policies that are issued and implemented simultaneously. This is crucial for effectively addressing the issue of diseases that transfer from animals to humans." (FGD 2)	Fragmented policymaking and a lack of coordination between agencies lead to misaligned indicators and ineffective responses to zoonotic diseases.	Develop and implement integrated cross-sectoral policies that align goals, indicators, and timelines across health, veterinary, and environmental agencies.	Health, animal and wildlife, and environmental government sectors collaborate to create unified policy frameworks under a One Health approach. Joint task forces and shared planning workshops ensure policies are launched and implemented in sync.	More effective prevention and control of zoonotic diseases, improved coordination and efficiency across agencies, clearer guidance for local authorities, and better protection of both human and animal health at the community level.
5	"From the perspective of those at the ground level, sometimes policies are created at higher levels without considering the realities of implementation. While the policy may	The central government is not fully aware of specific district or	Allow lower local levels to participate in policy making and allow flexibility	Provincial governors provide templates; districts	Interventions fit local ecology and culture, boosting

	seem straightforward to those above, it can be challenging for practitioners who need to coordinate with various departments.” (FGD 2) “There are discussions and debates, weighing economic concerns against health priorities and considering various aspects specific to each province. For instance, a tourist province with tourist attractions may view things differently and may find it challenging to implement certain policies. Each province and area has its unique characteristics, requiring consideration of these different perspectives, so it may not be feasible to implement disease surveillance, prevention, and control to the full 100% in line with policy.” (FGD 2) “If a local area wants to tackle disease, they can leverage these resources and handle it independently. They don't need to follow a standardized provincial policy, where the same solution applies across every district. For instance, if my village doesn't have the Nong Sai Ya issue, why should we implement that plan? Flexibility should be allowed.” (FGD 2)	subdistrict issues and characteristics, so there is a disconnect when policy is made.	when applying it to provinces with different contexts.	adapt; central audits outcomes, not methods	effectiveness & ownership
6	“If they (the Security Department) focus on securing border areas and preventing illegal entry, it would be a comprehensive measure.” (FGD 2)	Unsecured border areas allow illegal movement of people and animals	Securing border areas and preventing illegal entry	The Security Department establishes more checkpoints and utilizes technology to better control border regions	Fewer cross-border introductions of disease. Protects border communities, livestock, and national wildlife health
7	“In terms of public health, the focus will be on enforcing laws that already exist but are often not implemented, such as the Public Cleanliness Act, Public Health Act, and Disease Control Act. Proper enforcement of these laws could eradicate all diseases, not only those spread by animals to humans. For example, the Public Cleanliness Act regulates environmental conditions to keep homes and neighborhoods clean, reducing nuisances. Raising animals like	Existing public health laws (e.g., Public Cleanliness Act, Public Health Act, Disease Control Act) are not effectively enforced, leading to preventable health risks and	Strengthen the enforcement of existing public health laws and regulations related to hygiene, animal ownership, and environmental cleanliness.	Central, provincial, district, subdistrict, and village levels coordinate to enforce laws more consistently. This includes regular	Improved environmental, animal, and human health conditions for communities, with reduced incidence of communicable diseases (including zoonotic ones),

	pigs, dogs, or other pets requires registration and permission, making sure everything follows legal protocols. If these laws were strictly enforced, similar to other countries with strict cleanliness rules like prohibiting spitting, we could effectively prevent and control disease.” (FGD 2)	poor environmental hygiene.		inspections, penalties for violations, public awareness campaigns, and inter-agency collaboration to support compliance.	stronger public health systems, and cleaner and safer villages.
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Area 5: Potential interventions related to leveraging collaboration networks

Table 10 highlights the importance of leveraging collaboration networks for improving disease management, communication, and coordination. One potential intervention focuses on reducing delays in the use of scientific data by strengthening partnerships between research institutions and government agencies. Health, wildlife, and veterinary sectors are encouraged to establish collaborative frameworks that allow earlier access to research findings. This would accelerate informed decision-making and improve disease surveillance and prevention by ensuring that critical knowledge is not stuck in lengthy publication cycles.

Another major intervention addresses the breakdown of communication across different government levels. To resolve this, central and provincial authorities are advised to adopt modern information technology and systems that ensure clear and timely sharing with local units [98]. By decentralizing decision-making and streamlining communication through technology, local teams can receive direct instructions, act faster, and resolve issues more effectively. This empowers local governance, supports rapid responses to outbreaks, and ensures clearer accountability, ultimately benefiting both affected populations and broader economic stability.

Table 10. Potential interventions related to leveraging collaboration networks

#	Quote	Problem Identified	Measure/ Intervention Suggested	How (Who & How)	Impact (What & Who)
1	“Often, the information is tied up in research, which can take years to be published and become accessible. If we’re trying to set up a surveillance system, we should have access to this information more quickly. Sometimes, waiting 2–3 years for it to be published delays sharing this knowledge with government agencies or relevant personnel, which is critical.” (FGD 2)	Limited use of research in disease management practices due to the long publication process	Strengthening partnerships with research institutions so they can have early access to results and data	Health, wildlife, and veterinary government agencies establish research collaboration frameworks	Enhanced scientific knowledge and informed decision-making

2	"However, coordinating at the local level has challenges. For one, communication is often understood at higher levels, but by the time it reaches the on-ground operators, issues arise if the instructions aren't clear or haven't been passed along effectively. This can lead to local teams working in different directions, complicating area-based coordination. Take, for example, projects like avian flu or rabies. Letters are sent to the provincial governor, requesting collaboration across three or four agencies. But coordination problems arise because, while a letter reaches the top, the local units might not even know it exists. This is one real issue on the ground. Sometimes, when it finally reaches the local level, they're unsure if it's even part of their duties, like dealing with pets or stray animals, which is directly their responsibility." (FGD 2) "Actually, it might begin with the transfer of policy from the central level down to the provincial level. At the central level, there is a lot of planning happening, but by the time it reaches the provincial level, they often say they were never informed or never saw that there was an MOU in place. They never saw that this type of plan was required, so they don't act on it, thinking that no orders have been issued from above. Without clear directives, they don't feel confident to initiate anything, which makes it quite challenging to drive progress at the provincial level." (FGD 2) "It becomes a chain of individual units holding responsibility until it reaches the ministry level, then across ministries, which involves additional regulatory review. This whole process takes an enormous amount of time because of all the document handling and format requirements. For instance, by the time an official notice or warning about a disease reaches the public, the outbreak could be over. Even with "urgent" stamped on it, the entire process is far from being truly urgent." (FGD 2)	Lack of coordination at the local level due to miscommunication and misunderstandings arising from a disconnected top-down system, where there is a lack of communication between the central, provincial, and local levels. Slow hierarchical information flow delays outbreak response	Take a few steps into decentralization, adhocracy, and strengthen communication, using more modern channels so it's easier to keep everyone on the loop and so information doesn't get lost.	Central and provincial authorities use information technology or systems to share tasks clearly with local units.	Local teams get clear instructions, improving response and reducing miscommunication. Allows local areas to identify, address, and resolve their issues, thus empower local governance. Clear authority & accountability, faster local decisions, better service to citizens. Faster response, limited spread, protection for affected populations and the wider economy
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Main Conclusions

“Everyone is like a gear in a clock, right? Whether it's a small or large gear, whether it's a community member or a government agency, everything must work together, moving forward with mutual respect and importance. (...) No one is more or less significant; everyone has an equally essential role. This becomes a collective strength within the community, enabling everything to move forward.” – Stakeholder from FGD 1

Causal Relationship Diagrams

The causal relationship diagrams presented in this deliverable offer a systems-level understanding of the complex, interdependent factors influencing zoonotic spillover, disease outbreaks, and pandemic preparedness in rural, borderland, and ethnically diverse settings. Synthesizing across all models, several key themes emerge. Land use changes, such as deforestation, forest encroachment, and agricultural expansion, disrupt ecological balances and increase contact between humans, domestic animals, and wildlife. These pressures, compounded by climate-related events like floods and droughts, heighten the risk of zoonotic transmission by creating breeding grounds for disease vectors and contaminating shared natural resources. Human-wildlife coexistence tensions, if not managed inclusively, further destabilize ecosystems and contribute to spillover dynamics.

High levels of labor mobility, cross-border migration, and tourism accelerate the spread of endemic and emerging diseases. Undocumented migration and informal labor practices create barriers to health access and drive disease concealment due to fear of enforcement. Similarly, the movement of animals, legally and illegally, across borders or through informal trade channels introduces poorly regulated health risks. Weak health screening and fragmented coordination among institutions compound these vulnerabilities.

Traditional medical beliefs, low health literacy, and inconsistent health education, especially among ethnic minority communities, result in widespread engagement in high-risk behaviors. These include the consumption of wild or raw food, poor food handling, and raising unvaccinated stray or wild animals. Deeply held beliefs can discourage the adoption of modern preventive measures. Education gaps in both schools and community outreach weaken long-term behavioral change and pandemic readiness. The strategic use of information technology emerged as a cross-cutting enabler of improved health outcomes. IT enhances outbreak mapping, facilitates data sharing, enables effective health communication, and supports remote adaptation during isolation. However, the benefits of technology are contingent on its integration into institutional systems, interagency coordination, and public trust. Poor communication infrastructure or fragmented information flow can lead to duplication of efforts, wasted resources, and misinformation.

The models collectively highlight that effective disease prevention and response are not the result of isolated interventions, but of interconnected and reinforcing actions across environmental, social, institutional, and technological domains. Improving health education,

addressing systemic inequalities, strengthening animal and human health systems, and building community trust are all necessary components of a One Health approach. Furthermore, future preparedness depends on the ability to coordinate efforts, share data, and act on early signals, especially in vulnerable, mobile, and high-risk contexts.

Potential interventions

The potential interventions explored across the tables emphasize a holistic and multisectoral approach to improving public, animal, and environmental health. A recurring theme is the importance of empowering local actors, such as Village Health Volunteers (VHVs), district-level agencies, and community leaders, by enhancing their training, ensuring clarity in their roles, and integrating their insights into decision-making processes. Interventions call for local governments to play an active role in adapting health and disease prevention strategies to fit the cultural, social, and ecological context of their areas. When local actors are given ownership and supported with adequate resources and guidance, interventions become more sustainable, relevant, and responsive.

Equally vital is the strengthening of governance systems and legal frameworks to support health outcomes. Many challenges stem from laws and policies that are disconnected from local realities, inconsistently enforced, or overly rigid. Suggested solutions include co-developing laws with community participation, reviewing outdated or poorly designed policies, and fostering flexibility in national plans so that provinces and districts can tailor them to their specific contexts. Clearer protocols and more integrated planning are needed to avoid fragmented responses and to coordinate across sectors, such as public health, agriculture, wildlife, and security, especially for complex issues like zoonotic spillovers and disease outbreaks. Cross-border health threats further highlight the need for policies that are not only coherent but also enforced through modern tools and intergovernmental coordination.

Technology and research tools also play a crucial role in enhancing the effectiveness and timeliness of responses. Delayed access to scientific research, inconsistent information flow, and hierarchical communication breakdowns were all identified as barriers. Proposed solutions include strengthening partnerships between research institutions and government agencies to speed up data sharing and deploying modern information and communication technologies to ensure clear, rapid communication from national to local levels. These tools, when used strategically, improve decision-making, coordination, and transparency, allowing frontline responders to act confidently and efficiently during emergencies.

Limitations

We acknowledge several limitations in this work. This study is based on two focus group discussions conducted with a purposively selected sample of stakeholders from civil society and government agencies. The results reflect the views of participants with relevant experience with issues related to zoonotic spillover risk. The goal is to inform specific, context-sensitive interventions in Thailand's border regions.

The relationship diagrams presented in this report are based on participants' qualitative statements and describe associations between variables as perceived by stakeholders. These associations are not causal claims and should not be interpreted as mechanistic explanations. The diagrams are constructed from subjective input and depend on the accurate extraction and synthesis of qualitative data.

The analysis is limited to the thematic scope and stakeholder input collected during the FGDs. While efforts were made to include a range of institutional and community-level perspectives, the study does not represent all relevant actors involved in zoonotic spillover prevention. For example, perspectives from private sector actors, migrant worker communities, or local indigenous groups were not directly captured.

Finally, the outputs presented here are preliminary and will be further refined through upcoming stakeholder workshops. These workshops aim to validate relationship diagrams, assess the relevance of proposed interventions, and support the design of context-specific public health strategies.

Future Research

Future research could build upon the current model and intervention matrix by testing it and validating it in new contexts through qualitative investigations. Specifically, it would be valuable to conduct follow-up studies in provinces such as Chanthaburi and Chiang Rai, using FGDs or participatory approaches like group-based causal loop diagram (CLD) building. These methods would allow for a deeper exploration of local perceptions, feedback dynamics, and contextual adaptations of the intervention framework. Such comparative insights could strengthen the applicability of the model across diverse rural settings and inform more tailored, community-driven strategies for zoonotic spillover prevention.

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